Modeling Neutron Stars
Lecture-8

Chiranjib Mondal
LPC, Université Caen Normandie
https://sites.google.com/view/nucleartheory/home

UNIVERSITÉ CAEN NORMANDIE

Nuclear Theory Group
February 13, 2021

Contents:

Elementary Field Theory

- Recap: Infinite Nuclear matter
- Recap of $\sigma - \omega - \rho$ model
- Determination of coupling constants

Nuclear Matter

Microscopic description

- Energy per particle of nuclear matter of density $\rho_b = \rho_n + \rho_p$ and asymmetry $\delta = (\rho_n - \rho_p)/\rho_b$ is given by,

$$\frac{\epsilon}{\rho_b}(\rho_b, \delta) \simeq \frac{\epsilon}{\rho_b}(\rho_b, 0) + e_{\rm sym}(\rho_b)\delta^2; \quad e_{\rm sym}(\rho_b) = \frac{1}{2}\left(\frac{d^2\left(\frac{\epsilon}{\rho_b}\right)(\rho_b, \delta)}{d\delta^2}\right)_{\delta=0}.$$

Nuclear Matter

Microscopic description

- Energy per particle of nuclear matter of density $\rho_b = \rho_n + \rho_p$ and asymmetry $\delta = (\rho_n - \rho_p)/\rho_b$ is given by,

$$\frac{\epsilon}{\rho_b}(\rho_b, \delta) \simeq \frac{\epsilon}{\rho_b}(\rho_b, 0) + e_{\rm sym}(\rho_b)\delta^2; \quad e_{\rm sym}(\rho_b) = \frac{1}{2}\left(\frac{d^2\left(\frac{\epsilon}{\rho_b}\right)(\rho_b, \delta)}{d\delta^2}\right)_{\delta=0}.$$

$$\frac{\epsilon}{\rho_b}(\rho_b, 0) \simeq \left(\frac{\epsilon}{\rho_b}\right)_0 + \frac{1}{2}K_0\left(\frac{\rho_b - \rho_0}{3\rho_0}\right)^2; \quad K_0 = 9\rho_0^2\left(\frac{d^2\frac{\epsilon}{\rho_b}(\rho_b, 0)}{d\rho_b^2}\right)_{\rho_0}.$$

Nuclear Matter

Microscopic description

- Energy per particle of nuclear matter of density $\rho_b = \rho_n + \rho_p$ and asymmetry $\delta = (\rho_n - \rho_p)/\rho_b$ is given by,

$$\frac{\epsilon}{\rho_b}(\rho_b, \delta) \simeq \frac{\epsilon}{\rho_b}(\rho_b, 0) + e_{\rm sym}(\rho_b)\delta^2; \quad e_{\rm sym}(\rho_b) = \frac{1}{2}\left(\frac{d^2\left(\frac{\epsilon}{\rho_b}\right)(\rho_b, \delta)}{d\delta^2}\right)_{\delta=0}.$$

$$\frac{\epsilon}{\rho_b}(\rho_b, 0) \simeq \left(\frac{\epsilon}{\rho_b}\right)_0 + \frac{1}{2}K_0\left(\frac{\rho_b - \rho_0}{3\rho_0}\right)^2; \quad K_0 = 9\rho_0^2\left(\frac{d^2\frac{\epsilon}{\rho_b}(\rho_b, 0)}{d\rho_b^2}\right)_{\rho_0}.$$

- The symmetry energy is expanded as

$$e_{\rm sym}(\rho_b) \simeq J + L\left(\frac{\rho_b - \rho_0}{3\rho_0}\right) + \frac{1}{2}K_{\rm sym}\left(\frac{\rho_b - \rho_0}{3\rho_0}\right)^2$$

- Here, $J = e_{\rm sym}(\rho_0)$, $L = \left(3\rho\frac{de_{\rm sym}}{d\rho_b}\right)_{\rho_0}$, $K_{\rm sym} = 9\rho_0^2\left(\frac{d^2 e_{\rm sym}}{d\rho_b^2}\right)_{\rho_0}$.

Nuclear Matter

Microscopic description

- Energy per particle of nuclear matter of density $\rho_b = \rho_n + \rho_p$ and asymmetry $\delta = (\rho_n - \rho_p)/\rho_b$ is given by,

$$\frac{\epsilon}{\rho_b}(\rho_b, \delta) \simeq \frac{\epsilon}{\rho_b}(\rho_b, 0) + e_{\rm sym}(\rho_b)\delta^2; \quad e_{\rm sym}(\rho_b) = \frac{1}{2}\left(\frac{d^2\left(\frac{\epsilon}{\rho_b}\right)(\rho_b, \delta)}{d\delta^2}\right)_{\delta=0}.$$

$$\frac{\epsilon}{\rho_b}(\rho_b, 0) \simeq \left(\frac{\epsilon}{\rho_b}\right)_0 + \frac{1}{2}K_0\left(\frac{\rho_b - \rho_0}{3\rho_0}\right)^2; \quad K_0 = 9\rho_0^2\left(\frac{d^2\frac{\epsilon}{\rho_b}(\rho_b, 0)}{d\rho_b^2}\right)_{\rho_0}.$$

- The symmetry energy is expanded as

$$e_{\rm sym}(\rho_b) \simeq J + L\left(\frac{\rho_b - \rho_0}{3\rho_0}\right) + \frac{1}{2}K_{\rm sym}\left(\frac{\rho_b - \rho_0}{3\rho_0}\right)^2$$

- Here, $J = e_{\rm sym}(\rho_0)$, $L = \left(3\rho\frac{de_{\rm sym}}{d\rho_b}\right)_{\rho_0}$, $K_{\rm sym} = 9\rho_0^2\left(\frac{d^2 e_{\rm sym}}{d\rho_b^2}\right)_{\rho_0}$.
- The pressure is given by, $\mathtt{p}(\rho_b) = \rho_b^2\frac{d\left(\frac{\epsilon}{\rho_b}\right)(\rho_b, 0)}{d\rho_b}$.

Lagrangian and Field equations

$\sigma - \omega - \rho$ model

Lagrangian and Field equations

$\sigma - \omega - \boldsymbol{\rho}$ model

- The Lagrangian is given by

$$\Rightarrow \quad \mathscr{L} = \mathscr{L}_\sigma + \mathscr{L}_\omega + \mathscr{L}_{\boldsymbol{\rho}} + \mathscr{L}_\psi + \mathscr{L}_{int} - U(\sigma)$$

$$U(\sigma) = \frac{1}{3}\mathrm{g}_2\sigma^3 + \frac{1}{4}\mathrm{g}_3\sigma^4$$

$$\mathscr{L}_{int} = \mathrm{g}_\sigma\overline{\psi}\psi\sigma - \mathrm{g}_\omega\overline{\psi}\gamma^\mu\psi\omega_\mu - \frac{1}{2}\mathrm{g}_{\boldsymbol{\rho}}\overline{\psi}\gamma^\mu\boldsymbol{\tau}\psi\cdot\boldsymbol{\rho}_\mu.$$

Lagrangian and Field equations

$\sigma - \omega - \boldsymbol{\rho}$ model

- The Lagrangian is given by

$$\Rightarrow \quad \mathscr{L} = \mathscr{L}_\sigma + \mathscr{L}_\omega + \mathscr{L}_{\boldsymbol{\rho}} + \mathscr{L}_\psi + \mathscr{L}_{int} - U(\sigma)$$

$$U(\sigma) = \frac{1}{3}\mathrm{g}_2\sigma^3 + \frac{1}{4}\mathrm{g}_3\sigma^4$$

$$\mathscr{L}_{int} = \mathrm{g}_\sigma\overline{\psi}\psi\sigma - \mathrm{g}_\omega\overline{\psi}\gamma^\mu\psi\omega_\mu - \frac{1}{2}\mathrm{g}_{\boldsymbol{\rho}}\overline{\psi}\gamma^\mu\boldsymbol{\tau}\psi\cdot\boldsymbol{\rho}_\mu.$$

- The field equations under charge conservation and time reversal symmetry for static mean fields are given by

$$m_\sigma^2\sigma = \mathrm{g}_\sigma\rho_s - \mathrm{g}_2\sigma^2 - \mathrm{g}_3\sigma^3,$$

$$m_\omega^2\omega = \mathrm{g}_\omega\rho_b.$$

$$m_{\boldsymbol{\rho}}^2\rho_3^0 = \frac{\mathrm{g}_{\boldsymbol{\rho}}}{2}(\rho_n - \rho_p).$$

Lagrangian and Field equations

$\sigma - \omega - \boldsymbol{\rho}$ model

- The Lagrangian is given by

$$\Rightarrow \quad \mathscr{L} = \mathscr{L}_\sigma + \mathscr{L}_\omega + \mathscr{L}_{\boldsymbol{\rho}} + \mathscr{L}_\psi + \mathscr{L}_{int} - U(\sigma)$$

$$U(\sigma) = \frac{1}{3}\mathrm{g}_2\sigma^3 + \frac{1}{4}\mathrm{g}_3\sigma^4$$

$$\mathscr{L}_{int} = \mathrm{g}_\sigma\overline{\psi}\psi\sigma - \mathrm{g}_\omega\overline{\psi}\gamma^\mu\psi\omega_\mu - \frac{1}{2}\mathrm{g}_{\boldsymbol{\rho}}\overline{\psi}\gamma^\mu\boldsymbol{\tau}\psi\cdot\boldsymbol{\rho}_\mu.$$

- The field equations under charge conservation and time reversal symmetry for static mean fields are given by

$$m_\sigma^2\sigma = \mathrm{g}_\sigma\rho_s - \mathrm{g}_2\sigma^2 - \mathrm{g}_3\sigma^3,$$

$$m_\omega^2\omega = \mathrm{g}_\omega\rho_b.$$

$$m_{\boldsymbol{\rho}}^2\rho_3^0 = \frac{\mathrm{g}_{\boldsymbol{\rho}}}{2}(\rho_n - \rho_p).$$

Here, $\rho_b = \frac{2k_f^3}{3\pi^2}$ and $\rho_s = \frac{2}{\pi^2}\int_0^{k_f} p^2\,dp\frac{m-\mathrm{g}_\sigma\sigma}{\sqrt{p^2+(m-\mathrm{g}_\sigma\sigma)^2}}$ and $\rho_q = \rho_b\left(\frac{1\pm\delta}{2}\right)$.

Energy density and pressure

$\sigma - \omega - \rho$ model

Energy density and pressure

$\sigma - \omega - \boldsymbol{\rho}$ model

- Energy density of the $\sigma - \omega - \boldsymbol{\rho}$ model is

$$\begin{aligned}\epsilon &= \frac{1}{2}m_\sigma^2\sigma^2 + \frac{1}{3}g_2\sigma^3 + \frac{1}{4}g_3\sigma^4 + \frac{1}{2}m_\omega^2\omega_0^2 + \frac{1}{2}m_{\boldsymbol{\rho}}^2\rho_{0,3}^2 \\ &+ \frac{1}{\pi^2}\left[\int_0^{k_n} p^2\,dp\sqrt{p^2+m^{*2}} + \int_0^{k_p} p^2\,dp\sqrt{p^2+m^{*2}}\right]\end{aligned}$$

Energy density and pressure

$\sigma - \omega - \boldsymbol{\rho}$ model

- Energy density of the $\sigma - \omega - \boldsymbol{\rho}$ model is

$$\begin{aligned}\epsilon &= \frac{1}{2}m_\sigma^2\sigma^2 + \frac{1}{3}g_2\sigma^3 + \frac{1}{4}g_3\sigma^4 + \frac{1}{2}m_\omega^2\omega_0^2 + \frac{1}{2}m_{\boldsymbol{\rho}}^2\rho_{0,3}^2 \\ &+ \frac{1}{\pi^2}\left[\int_0^{k_n} p^2\,dp\sqrt{p^2+m^{*2}} + \int_0^{k_p} p^2\,dp\sqrt{p^2+m^{*2}}\right]\end{aligned}$$

- Pressure $\mathtt{p}$ of the $\sigma - \omega - \boldsymbol{\rho}$ model is written as

$$\begin{aligned}\mathtt{p} &= -\frac{1}{2}m_\sigma^2\sigma^2 - \frac{1}{3}g_2\sigma^3 - \frac{1}{4}g_3\sigma^4 + \frac{1}{2}m_\omega^2\omega_0^2 + \frac{1}{2}m_{\boldsymbol{\rho}}^2\rho_{0,3}^2 \\ &+ \frac{1}{3\pi^2}\left[\int_0^{k_n} p^2\,dp\frac{dp^2}{\sqrt{p^2+m^{*2}}} + \int_0^{k_p} p^2\,dp\frac{dp^2}{\sqrt{p^2+m^{*2}}}\right]\end{aligned}$$

Recalibration of symmetric matter

$\sigma - \omega - \boldsymbol{\rho}$ model

Recalibration of symmetric matter

$\sigma - \omega - \boldsymbol{\rho}$ model

- For symmetric matter, $\boldsymbol{\rho}$ meson does not contribute (see $\boldsymbol{\rho}$ field equation).

Recalibration of symmetric matter

$\sigma - \omega - \boldsymbol{\rho}$ model

- For symmetric matter, $\boldsymbol{\rho}$ meson does not contribute (see $\boldsymbol{\rho}$ field equation).
- We define,

$$\Phi = \mathrm{g}_\sigma \sigma, \;\; W = \mathrm{g}_\omega \omega, \;\; c_\sigma = \frac{\mathrm{g}_\sigma}{m_\sigma}, \;\; c_\omega = \frac{\mathrm{g}_\omega}{m_\omega}, \;\; \overline{\mathrm{g}_2} = \frac{\mathrm{g}_2}{\mathrm{g}_\sigma^3}, \;\; \overline{\mathrm{g}_3} = \frac{\mathrm{g}_3}{\mathrm{g}_\sigma^4}.$$

Recalibration of symmetric matter

$\sigma - \omega - \boldsymbol{\rho}$ model

- For symmetric matter, $\boldsymbol{\rho}$ meson does not contribute (see $\boldsymbol{\rho}$ field equation).
- We define,

$$\Phi = \mathrm{g}_\sigma \sigma, \;\; W = \mathrm{g}_\omega \omega, \;\; c_\sigma = \frac{\mathrm{g}_\sigma}{m_\sigma}, \;\; c_\omega = \frac{\mathrm{g}_\omega}{m_\omega}, \;\; \overline{\mathrm{g}_2} = \frac{\mathrm{g}_2}{\mathrm{g}_\sigma^3}, \;\; \overline{\mathrm{g}_3} = \frac{\mathrm{g}_3}{\mathrm{g}_\sigma^4}.$$

- Energy density of the $\sigma - \omega - \boldsymbol{\rho}$ model is

$$\epsilon = \frac{1}{2} m_\sigma^2 \sigma^2 + \frac{1}{3} \mathrm{g}_2 \sigma^3 + \frac{1}{4} \mathrm{g}_3 \sigma^4 + \cdots$$

Recalibration of symmetric matter

$\sigma - \omega - \boldsymbol{\rho}$ model

- For symmetric matter, $\boldsymbol{\rho}$ meson does not contribute (see $\boldsymbol{\rho}$ field equation).
- We define,

$$\Phi = \mathrm{g}_\sigma \sigma, \;\; W = \mathrm{g}_\omega \omega, \;\; c_\sigma = \frac{\mathrm{g}_\sigma}{m_\sigma}, \;\; c_\omega = \frac{\mathrm{g}_\omega}{m_\omega}, \;\; \overline{\mathrm{g}_2} = \frac{\mathrm{g}_2}{\mathrm{g}_\sigma^3}, \;\; \overline{\mathrm{g}_3} = \frac{\mathrm{g}_3}{\mathrm{g}_\sigma^4}.$$

- Energy density of the $\sigma - \omega - \boldsymbol{\rho}$ model is

$$\epsilon = \frac{1}{2} m_\sigma^2 \sigma^2 + \frac{1}{3} \mathrm{g}_2 \sigma^3 + \frac{1}{4} \mathrm{g}_3 \sigma^4 + \cdots = \frac{1}{2} \frac{\Phi^2}{c_\sigma^2} + \frac{1}{3} \overline{\mathrm{g}_2} \Phi^3 + \frac{1}{4} \overline{\mathrm{g}_3} \Phi^4 + \cdots$$

Recalibration of symmetric matter

$\sigma - \omega - \boldsymbol{\rho}$ model

- For symmetric matter, $\boldsymbol{\rho}$ meson does not contribute (see $\boldsymbol{\rho}$ field equation).
- We define,

$$\Phi = \mathrm{g}_\sigma \sigma, \;\; W = \mathrm{g}_\omega \omega, \;\; c_\sigma = \frac{\mathrm{g}_\sigma}{m_\sigma}, \;\; c_\omega = \frac{\mathrm{g}_\omega}{m_\omega}, \;\; \overline{\mathrm{g}_2} = \frac{\mathrm{g}_2}{\mathrm{g}_\sigma^3}, \;\; \overline{\mathrm{g}_3} = \frac{\mathrm{g}_3}{\mathrm{g}_\sigma^4}.$$

- Energy density of the $\sigma - \omega - \boldsymbol{\rho}$ model is

$$\epsilon = \frac{1}{2} m_\sigma^2 \sigma^2 + \frac{1}{3} \mathrm{g}_2 \sigma^3 + \frac{1}{4} \mathrm{g}_3 \sigma^4 + \cdots = \frac{1}{2} \frac{\Phi^2}{c_\sigma^2} + \frac{1}{3} \overline{\mathrm{g}_2} \Phi^3 + \frac{1}{4} \overline{\mathrm{g}_3} \Phi^4 + \cdots$$

- We define

$$V(\Phi) \quad = \quad \frac{1}{2} \frac{\Phi^2}{c_\sigma^2} + \frac{1}{3} \overline{\mathrm{g}_2} \Phi^3 + \frac{1}{4} \overline{\mathrm{g}_3} \Phi^4$$

Recalibration of symmetric matter

$\sigma - \omega - \boldsymbol{\rho}$ model

- For symmetric matter, $\boldsymbol{\rho}$ meson does not contribute (see $\boldsymbol{\rho}$ field equation).
- We define,

$$\Phi = \mathrm{g}_\sigma \sigma, \;\; W = \mathrm{g}_\omega \omega, \;\; c_\sigma = \frac{\mathrm{g}_\sigma}{m_\sigma}, \;\; c_\omega = \frac{\mathrm{g}_\omega}{m_\omega}, \;\; \overline{\mathrm{g}_2} = \frac{\mathrm{g}_2}{\mathrm{g}_\sigma^3}, \;\; \overline{\mathrm{g}_3} = \frac{\mathrm{g}_3}{\mathrm{g}_\sigma^4}.$$

- Energy density of the $\sigma - \omega - \boldsymbol{\rho}$ model is

$$\epsilon = \frac{1}{2} m_\sigma^2 \sigma^2 + \frac{1}{3} \mathrm{g}_2 \sigma^3 + \frac{1}{4} \mathrm{g}_3 \sigma^4 + \cdots = \frac{1}{2} \frac{\Phi^2}{c_\sigma^2} + \frac{1}{3} \overline{\mathrm{g}_2} \Phi^3 + \frac{1}{4} \overline{\mathrm{g}_3} \Phi^4 + \cdots$$

- We define

$$\begin{aligned} V(\Phi) &= \frac{1}{2} \frac{\Phi^2}{c_\sigma^2} + \frac{1}{3} \overline{\mathrm{g}_2} \Phi^3 + \frac{1}{4} \overline{\mathrm{g}_3} \Phi^4 \\ \Rightarrow V'(\Phi) &\equiv \frac{\partial V(\Phi)}{\partial \Phi} = \frac{1}{c_\sigma^2} \Phi + \overline{\mathrm{g}_2} \Phi^2 + \overline{\mathrm{g}_3} \Phi^3 \end{aligned}$$

Recalibration of symmetric matter

$\sigma - \omega - \boldsymbol{\rho}$ model

- For symmetric matter, $\boldsymbol{\rho}$ meson does not contribute (see $\boldsymbol{\rho}$ field equation).
- We define,

$$\Phi = \mathrm{g}_\sigma \sigma, \;\; W = \mathrm{g}_\omega \omega, \;\; c_\sigma = \frac{\mathrm{g}_\sigma}{m_\sigma}, \;\; c_\omega = \frac{\mathrm{g}_\omega}{m_\omega}, \;\; \overline{\mathrm{g}_2} = \frac{\mathrm{g}_2}{\mathrm{g}_\sigma^3}, \;\; \overline{\mathrm{g}_3} = \frac{\mathrm{g}_3}{\mathrm{g}_\sigma^4}.$$

- Energy density of the $\sigma - \omega - \boldsymbol{\rho}$ model is

$$\epsilon = \frac{1}{2} m_\sigma^2 \sigma^2 + \frac{1}{3} \mathrm{g}_2 \sigma^3 + \frac{1}{4} \mathrm{g}_3 \sigma^4 + \cdots = \frac{1}{2} \frac{\Phi^2}{c_\sigma^2} + \frac{1}{3} \overline{\mathrm{g}_2} \Phi^3 + \frac{1}{4} \overline{\mathrm{g}_3} \Phi^4 + \cdots$$

- We define

$$\begin{aligned} V(\Phi) &= \frac{1}{2} \frac{\Phi^2}{c_\sigma^2} + \frac{1}{3} \overline{\mathrm{g}_2} \Phi^3 + \frac{1}{4} \overline{\mathrm{g}_3} \Phi^4 \\ \Rightarrow V'(\Phi) &\equiv \frac{\partial V(\Phi)}{\partial \Phi} = \frac{1}{c_\sigma^2} \Phi + \overline{\mathrm{g}_2} \Phi^2 + \overline{\mathrm{g}_3} \Phi^3 \\ &= \frac{m_\sigma^2}{\mathrm{g}_\sigma} \sigma + \frac{\mathrm{g}_2}{\mathrm{g}_\sigma} \sigma^2 + \frac{\mathrm{g}_3}{\mathrm{g}_\sigma} \sigma^3 \end{aligned}$$

Recalibration of symmetric matter

$\sigma - \omega - \boldsymbol{\rho}$ model

- For symmetric matter, $\boldsymbol{\rho}$ meson does not contribute (see $\boldsymbol{\rho}$ field equation).
- We define,

$$\Phi = g_\sigma \sigma, \;\; W = g_\omega \omega, \;\; c_\sigma = \frac{g_\sigma}{m_\sigma}, \;\; c_\omega = \frac{g_\omega}{m_\omega}, \;\; \overline{g_2} = \frac{g_2}{g_\sigma^3}, \;\; \overline{g_3} = \frac{g_3}{g_\sigma^4}.$$

- Energy density of the $\sigma - \omega - \boldsymbol{\rho}$ model is

$$\epsilon = \frac{1}{2} m_\sigma^2 \sigma^2 + \frac{1}{3} g_2 \sigma^3 + \frac{1}{4} g_3 \sigma^4 + \cdots = \frac{1}{2} \frac{\Phi^2}{c_\sigma^2} + \frac{1}{3} \overline{g_2} \Phi^3 + \frac{1}{4} \overline{g_3} \Phi^4 + \cdots$$

- We define

$$\begin{aligned} V(\Phi) &= \frac{1}{2} \frac{\Phi^2}{c_\sigma^2} + \frac{1}{3} \overline{g_2} \Phi^3 + \frac{1}{4} \overline{g_3} \Phi^4 \\ \Rightarrow V'(\Phi) &\equiv \frac{\partial V(\Phi)}{\partial \Phi} = \frac{1}{c_\sigma^2} \Phi + \overline{g_2} \Phi^2 + \overline{g_3} \Phi^3 \\ &= \frac{m_\sigma^2}{g_\sigma} \sigma + \frac{g_2}{g_\sigma} \sigma^2 + \frac{g_3}{g_\sigma} \sigma^3 = \rho_s \text{ (using } \sigma \text{ field eq.)} \end{aligned}$$

As, $m_\sigma^2 \sigma = g_\sigma \rho_s - g_2 \sigma^2 - g_3 \sigma^3$.

Recalibration of symmetric matter

$\sigma - \omega - \boldsymbol{\rho}$ model

- For symmetric matter, $\boldsymbol{\rho}$ meson does not contribute (see $\boldsymbol{\rho}$ field equation).
- We define,

$$\Phi = \mathrm{g}_\sigma \sigma, \;\; W = \mathrm{g}_\omega \omega, \;\; c_\sigma = \frac{\mathrm{g}_\sigma}{m_\sigma}, \;\; c_\omega = \frac{\mathrm{g}_\omega}{m_\omega}, \;\; \overline{\mathrm{g}_2} = \frac{\mathrm{g}_2}{\mathrm{g}_\sigma^3}, \;\; \overline{\mathrm{g}_3} = \frac{\mathrm{g}_3}{\mathrm{g}_\sigma^4}.$$

- Energy density of the $\sigma - \omega - \boldsymbol{\rho}$ model is

$$\epsilon = \frac{1}{2} m_\sigma^2 \sigma^2 + \frac{1}{3} \mathrm{g}_2 \sigma^3 + \frac{1}{4} \mathrm{g}_3 \sigma^4 + \cdots = \frac{1}{2} \frac{\Phi^2}{c_\sigma^2} + \frac{1}{3} \overline{\mathrm{g}_2} \Phi^3 + \frac{1}{4} \overline{\mathrm{g}_3} \Phi^4 + \cdots$$

- We define

$$\begin{aligned} V(\Phi) &= \frac{1}{2} \frac{\Phi^2}{c_\sigma^2} + \frac{1}{3} \overline{\mathrm{g}_2} \Phi^3 + \frac{1}{4} \overline{\mathrm{g}_3} \Phi^4 \\ \Rightarrow V'(\Phi) &\equiv \frac{\partial V(\Phi)}{\partial \Phi} = \frac{1}{c_\sigma^2} \Phi + \overline{\mathrm{g}_2} \Phi^2 + \overline{\mathrm{g}_3} \Phi^3 \\ &= \frac{m_\sigma^2}{\mathrm{g}_\sigma} \sigma + \frac{\mathrm{g}_2}{\mathrm{g}_\sigma} \sigma^2 + \frac{\mathrm{g}_3}{\mathrm{g}_\sigma} \sigma^3 = \rho_s \text{ (using } \sigma \text{ field eq.)} \end{aligned}$$

As, $m_\sigma^2 \sigma = \mathrm{g}_\sigma \rho_s - \mathrm{g}_2 \sigma^2 - \mathrm{g}_3 \sigma^3$. So we have, $\boxed{V'(\Phi) = \rho_s}$.

Energy of symmetric matter

$\sigma - \omega - \boldsymbol{\rho}$ model

Energy of symmetric matter

$\sigma - \omega - \boldsymbol{\rho}$ model

- For symmetric matter, the recalibrated energy is written as,

$$\epsilon = \langle E \rangle_{k_f} + V(\Phi) + \frac{1}{2} m_\omega^2 \omega_0^2,$$

Energy of symmetric matter

$\sigma - \omega - \rho$ model

- For symmetric matter, the recalibrated energy is written as,

$$\epsilon = \langle E \rangle_{k_f} + V(\Phi) + \frac{1}{2} m_\omega^2 \omega_0^2, \text{ with, } \langle E \rangle_{k_f} = \frac{2}{\pi^2} \int_0^{k_f} p^2 \, dp \sqrt{p^2 + (m - \mathrm{g}_\sigma \sigma)^2}$$

Energy of symmetric matter

$\sigma - \omega - \rho$ model

- For symmetric matter, the recalibrated energy is written as,

$$\begin{aligned}\epsilon &= \langle E\rangle_{k_f} + V(\Phi) + \frac{1}{2}m_\omega^2\omega_0^2, \text{ with, } \langle E\rangle_{k_f} = \frac{2}{\pi^2}\int_0^{k_f} p^2\,dp\sqrt{p^2 + (m - \mathrm{g}_\sigma\sigma)^2} \\ &= \langle E\rangle_{k_f} + V(\Phi) + \frac{1}{2}\frac{W^2}{c_\omega^2},\end{aligned}$$

Energy of symmetric matter

$\sigma - \omega - \rho$ model

- For symmetric matter, the recalibrated energy is written as,

$$\begin{aligned}\epsilon &= \langle E\rangle_{k_f} + V(\Phi) + \frac{1}{2}m_\omega^2\omega_0^2, \text{ with, } \langle E\rangle_{k_f} = \frac{2}{\pi^2}\int_0^{k_f} p^2\,dp\sqrt{p^2 + (m - \mathrm{g}_\sigma\sigma)^2} \\ &= \langle E\rangle_{k_f} + V(\Phi) + \frac{1}{2}\frac{W^2}{c_\omega^2}, \text{ with, } \langle E\rangle_{k_f} = \frac{2}{\pi^2}\int_0^{k_f} p^2\,dp\sqrt{p^2 + (m - \Phi)^2}.\end{aligned}$$

Energy of symmetric matter

$\sigma - \omega - \rho$ model

- For symmetric matter, the recalibrated energy is written as,

$$\begin{aligned}
\epsilon &= \langle E \rangle_{k_f} + V(\Phi) + \frac{1}{2} m_\omega^2 \omega_0^2, \text{ with, } \langle E \rangle_{k_f} = \frac{2}{\pi^2} \int_0^{k_f} p^2 \, dp \sqrt{p^2 + (m - g_\sigma \sigma)^2} \\
&= \langle E \rangle_{k_f} + V(\Phi) + \frac{1}{2} \frac{W^2}{c_\omega^2}, \text{ with, } \langle E \rangle_{k_f} = \frac{2}{\pi^2} \int_0^{k_f} p^2 \, dp \sqrt{p^2 + (m - \Phi)^2}. \\
&= \langle E \rangle_{k_f} + V(\Phi) + \frac{1}{2} c_\omega^2 \rho_b^2,
\end{aligned}$$

Energy of symmetric matter

$\sigma - \omega - \rho$ model

- For symmetric matter, the recalibrated energy is written as,

$$\begin{aligned}
\epsilon &= \langle E \rangle_{k_f} + V(\Phi) + \frac{1}{2} m_\omega^2 \omega_0^2, \text{ with, } \langle E \rangle_{k_f} = \frac{2}{\pi^2} \int_0^{k_f} p^2 \, dp \sqrt{p^2 + (m - \mathrm{g}_\sigma \sigma)^2} \\
&= \langle E \rangle_{k_f} + V(\Phi) + \frac{1}{2} \frac{W^2}{c_\omega^2}, \text{ with, } \langle E \rangle_{k_f} = \frac{2}{\pi^2} \int_0^{k_f} p^2 \, dp \sqrt{p^2 + (m - \Phi)^2}. \\
&= \langle E \rangle_{k_f} + V(\Phi) + \frac{1}{2} c_\omega^2 \rho_b^2, \text{ as, } m_\omega^2 \omega = \mathrm{g}_\omega \rho_b \Rightarrow W = c_\omega^2 \rho_b
\end{aligned}$$

Energy of symmetric matter

$\sigma - \omega - \rho$ model

- For symmetric matter, the recalibrated energy is written as,

$$\begin{aligned}
\epsilon &= \langle E \rangle_{k_f} + V(\Phi) + \frac{1}{2} m_\omega^2 \omega_0^2, \text{ with, } \langle E \rangle_{k_f} = \frac{2}{\pi^2} \int_0^{k_f} p^2 \, dp \sqrt{p^2 + (m - \mathrm{g}_\sigma \sigma)^2} \\
&= \langle E \rangle_{k_f} + V(\Phi) + \frac{1}{2} \frac{W^2}{c_\omega^2}, \text{ with, } \langle E \rangle_{k_f} = \frac{2}{\pi^2} \int_0^{k_f} p^2 \, dp \sqrt{p^2 + (m - \Phi)^2}. \\
&= \langle E \rangle_{k_f} + V(\Phi) + \frac{1}{2} c_\omega^2 \rho_b^2, \text{ as, } m_\omega^2 \omega = \mathrm{g}_\omega \rho_b \Rightarrow W = c_\omega^2 \rho_b
\end{aligned}$$

- Now, the pressure at saturation vanishes. So we have,

$$\left[\frac{d}{d\rho_b} \left(\frac{\epsilon}{\rho_b} \right) \right]_{\rho_0} = 0$$

Energy of symmetric matter

$\sigma - \omega - \rho$ model

- For symmetric matter, the recalibrated energy is written as,

$$\begin{aligned}
\epsilon &= \langle E \rangle_{k_f} + V(\Phi) + \frac{1}{2} m_\omega^2 \omega_0^2, \text{ with, } \langle E \rangle_{k_f} = \frac{2}{\pi^2} \int_0^{k_f} p^2 \, dp \sqrt{p^2 + (m - \mathrm{g}_\sigma \sigma)^2} \\
&= \langle E \rangle_{k_f} + V(\Phi) + \frac{1}{2} \frac{W^2}{c_\omega^2}, \text{ with, } \langle E \rangle_{k_f} = \frac{2}{\pi^2} \int_0^{k_f} p^2 \, dp \sqrt{p^2 + (m - \Phi)^2}. \\
&= \langle E \rangle_{k_f} + V(\Phi) + \frac{1}{2} c_\omega^2 \rho_b^2, \text{ as, } m_\omega^2 \omega = \mathrm{g}_\omega \rho_b \Rightarrow W = c_\omega^2 \rho_b
\end{aligned}$$

- Now, the pressure at saturation vanishes. So we have,

$$\left[\frac{d}{d\rho_b} \left(\frac{\epsilon}{\rho_b} \right) \right]_{\rho_0} = 0 = \left[\left(\frac{\partial}{\partial \rho_b} + \frac{\partial \Phi}{\partial \rho_b} \frac{\partial}{\partial \Phi} \right) \left(\frac{\epsilon}{\rho_b} \right) \right]_{\rho_0}.$$

Energy of symmetric matter

$\sigma - \omega - \rho$ model

- For symmetric matter, the recalibrated energy is written as,

$$\begin{aligned}
\epsilon &= \langle E\rangle_{k_f} + V(\Phi) + \frac{1}{2}m_\omega^2\omega_0^2, \text{ with, } \langle E\rangle_{k_f} = \frac{2}{\pi^2}\int_0^{k_f} p^2\,dp\sqrt{p^2 + (m - \mathrm{g}_\sigma\sigma)^2} \\
&= \langle E\rangle_{k_f} + V(\Phi) + \frac{1}{2}\frac{W^2}{c_\omega^2}, \text{ with, } \langle E\rangle_{k_f} = \frac{2}{\pi^2}\int_0^{k_f} p^2\,dp\sqrt{p^2 + (m - \Phi)^2}. \\
&= \langle E\rangle_{k_f} + V(\Phi) + \frac{1}{2}c_\omega^2\rho_b^2, \text{ as, } m_\omega^2\omega = \mathrm{g}_\omega\rho_b \Rightarrow W = c_\omega^2\rho_b
\end{aligned}$$

- Now, the pressure at saturation vanishes. So we have,

$$\left[\frac{d}{d\rho_b}\left(\frac{\epsilon}{\rho_b}\right)\right]_{\rho_0} = 0 = \left[\left(\frac{\partial}{\partial\rho_b} + \frac{\partial\Phi}{\partial\rho_b}\frac{\partial}{\partial\Phi}\right)\left(\frac{\epsilon}{\rho_b}\right)\right]_{\rho_0}.$$

- Now simplify the second term,

$$\frac{\partial\epsilon}{\partial\Phi} = \frac{\partial\langle E\rangle_{k_f}}{\partial\Phi} + V'(\Phi)$$

Energy of symmetric matter

$\sigma - \omega - \rho$ model

- For symmetric matter, the recalibrated energy is written as,

$$\begin{aligned}
\epsilon &= \langle E\rangle_{k_f} + V(\Phi) + \frac{1}{2}m_\omega^2\omega_0^2, \text{ with, } \langle E\rangle_{k_f} = \frac{2}{\pi^2}\int_0^{k_f} p^2\,dp\sqrt{p^2+(m-\mathrm{g}_\sigma\sigma)^2} \\
&= \langle E\rangle_{k_f} + V(\Phi) + \frac{1}{2}\frac{W^2}{c_\omega^2}, \text{ with, } \langle E\rangle_{k_f} = \frac{2}{\pi^2}\int_0^{k_f} p^2\,dp\sqrt{p^2+(m-\Phi)^2}. \\
&= \langle E\rangle_{k_f} + V(\Phi) + \frac{1}{2}c_\omega^2\rho_b^2, \text{ as, } m_\omega^2\omega = \mathrm{g}_\omega\rho_b \Rightarrow W = c_\omega^2\rho_b
\end{aligned}$$

- Now, the pressure at saturation vanishes. So we have,

$$\left[\frac{d}{d\rho_b}\left(\frac{\epsilon}{\rho_b}\right)\right]_{\rho_0} = 0 = \left[\left(\frac{\partial}{\partial\rho_b} + \frac{\partial\Phi}{\partial\rho_b}\frac{\partial}{\partial\Phi}\right)\left(\frac{\epsilon}{\rho_b}\right)\right]_{\rho_0}.$$

- Now simplify the second term,

$$\frac{\partial\epsilon}{\partial\Phi} = \frac{\partial\langle E\rangle_{k_f}}{\partial\Phi} + V'(\Phi) = -\frac{2}{\pi^2}\int_0^{k_f} p^2\,dp\frac{(m-\Phi)}{\sqrt{p^2+(m-\Phi)^2}} + V'(\Phi)$$

Energy of symmetric matter

$\sigma - \omega - \rho$ model

- For symmetric matter, the recalibrated energy is written as,

$$\begin{aligned}
\epsilon &= \langle E\rangle_{k_f} + V(\Phi) + \frac{1}{2}m_\omega^2\omega_0^2, \text{ with, } \langle E\rangle_{k_f} = \frac{2}{\pi^2}\int_0^{k_f} p^2\,dp\sqrt{p^2+(m-\mathrm{g}_\sigma\sigma)^2} \\
&= \langle E\rangle_{k_f} + V(\Phi) + \frac{1}{2}\frac{W^2}{c_\omega^2}, \text{ with, } \langle E\rangle_{k_f} = \frac{2}{\pi^2}\int_0^{k_f} p^2\,dp\sqrt{p^2+(m-\Phi)^2}. \\
&= \langle E\rangle_{k_f} + V(\Phi) + \frac{1}{2}c_\omega^2\rho_b^2, \text{ as, } m_\omega^2\omega = \mathrm{g}_\omega\rho_b \Rightarrow W = c_\omega^2\rho_b
\end{aligned}$$

- Now, the pressure at saturation vanishes. So we have,

$$\left[\frac{d}{d\rho_b}\left(\frac{\epsilon}{\rho_b}\right)\right]_{\rho_0} = 0 = \left[\left(\frac{\partial}{\partial\rho_b} + \frac{\partial\Phi}{\partial\rho_b}\frac{\partial}{\partial\Phi}\right)\left(\frac{\epsilon}{\rho_b}\right)\right]_{\rho_0}.$$

- Now simplify the second term,

$$\begin{aligned}
\frac{\partial\epsilon}{\partial\Phi} = \frac{\partial\langle E\rangle_{k_f}}{\partial\Phi} + V'(\Phi) &= -\frac{2}{\pi^2}\int_0^{k_f} p^2\,dp\frac{(m-\Phi)}{\sqrt{p^2+(m-\Phi)^2}} + V'(\Phi) \\
&= -\rho_s + \rho_s = 0.
\end{aligned}$$

Determination of c_ω

$\sigma - \omega - \rho$ model

Determination of c_ω

$\sigma - \omega - \rho$ model

- So we write,

$$\left[\frac{d}{d\rho_b}\left(\frac{\epsilon}{\rho_b}\right)\right]_{\rho_0} = 0$$

Determination of c_ω

$\sigma - \omega - \rho$ model

- So we write,

$$\left[\frac{d}{d\rho_b}\left(\frac{\epsilon}{\rho_b}\right)\right]_{\rho_0} = 0 = \left[\frac{\partial}{\partial\rho_b}\left(\frac{\epsilon}{\rho_b}\right)\right]_{\rho_0}$$

Determination of c_ω

$\sigma - \omega - \rho$ model

- So we write,

$$\left[\frac{d}{d\rho_b}\left(\frac{\epsilon}{\rho_b}\right)\right]_{\rho_0} = 0 = \left[\frac{\partial}{\partial\rho_b}\left(\frac{\epsilon}{\rho_b}\right)\right]_{\rho_0} = \left[\frac{\partial\epsilon}{\partial\rho_b}\Big/\rho_b - \frac{\epsilon}{\rho_b^2}\right]_{\rho_0}$$

Determination of c_ω

$\sigma - \omega - \rho$ model

- So we write,

$$\left[\frac{d}{d\rho_b}\left(\frac{\epsilon}{\rho_b}\right)\right]_{\rho_0} = 0 = \left[\frac{\partial}{\partial\rho_b}\left(\frac{\epsilon}{\rho_b}\right)\right]_{\rho_0} = \left[\frac{\partial\epsilon}{\partial\rho_b}\Big/\rho_b - \frac{\epsilon}{\rho_b^2}\right]_{\rho_0}$$

$$\Rightarrow \quad \left(\frac{\partial\epsilon}{\partial\rho_b}\right)_{\rho_0} = \left(\frac{\epsilon}{\rho_b}\right)_{\rho_0}$$

Determination of c_ω

$\sigma - \omega - \rho$ model

- So we write,

$$\left[\frac{d}{d\rho_b}\left(\frac{\epsilon}{\rho_b}\right)\right]_{\rho_0} = 0 = \left[\frac{\partial}{\partial\rho_b}\left(\frac{\epsilon}{\rho_b}\right)\right]_{\rho_0} = \left[\frac{\partial\epsilon}{\partial\rho_b}\Big/\rho_b - \frac{\epsilon}{\rho_b^2}\right]_{\rho_0}$$

$$\Rightarrow \quad \left(\frac{\partial\epsilon}{\partial\rho_b}\right)_{\rho_0} = \left(\frac{\epsilon}{\rho_b}\right)_{\rho_0} = \left(\frac{B}{A}\right)_0 + m.$$

Determination of c_ω

$\sigma - \omega - \rho$ model

- So we write,

$$\left[\frac{d}{d\rho_b}\left(\frac{\epsilon}{\rho_b}\right)\right]_{\rho_0} = 0 = \left[\frac{\partial}{\partial\rho_b}\left(\frac{\epsilon}{\rho_b}\right)\right]_{\rho_0} = \left[\frac{\partial\epsilon}{\partial\rho_b}\Big/\rho_b - \frac{\epsilon}{\rho_b^2}\right]_{\rho_0}$$

$$\Rightarrow \quad \left(\frac{\partial\epsilon}{\partial\rho_b}\right)_{\rho_0} = \left(\frac{\epsilon}{\rho_b}\right)_{\rho_0} = \left(\frac{B}{A}\right)_0 + m.$$

- Simplifying the derivative,

$$\frac{\partial\epsilon}{\partial\rho_b} = \frac{\partial\langle E\rangle_{k_f}}{\partial\rho_b} + c_\omega^2\rho_b,$$

Determination of c_ω

$\sigma - \omega - \rho$ model

- So we write,

$$\left[\frac{d}{d\rho_b}\left(\frac{\epsilon}{\rho_b}\right)\right]_{\rho_0} = 0 = \left[\frac{\partial}{\partial\rho_b}\left(\frac{\epsilon}{\rho_b}\right)\right]_{\rho_0} = \left[\frac{\partial\epsilon}{\partial\rho_b}\Big/\rho_b - \frac{\epsilon}{\rho_b^2}\right]_{\rho_0}$$

$$\Rightarrow \quad \left(\frac{\partial\epsilon}{\partial\rho_b}\right)_{\rho_0} = \left(\frac{\epsilon}{\rho_b}\right)_{\rho_0} = \left(\frac{B}{A}\right)_0 + m.$$

- Simplifying the derivative,

$$\frac{\partial\epsilon}{\partial\rho_b} = \frac{\partial\langle E\rangle_{k_f}}{\partial\rho_b} + c_\omega^2\rho_b, \text{ with, } \langle E\rangle_{k_f} = \frac{2}{\pi^2}\int_0^{k_f} p^2\,dp\sqrt{p^2+m^{*2}} \;\&\; \rho_b = \frac{2k_f^3}{3\pi^2}.$$

Determination of c_ω

$\sigma - \omega - \rho$ model

- So we write,

$$\left[\frac{d}{d\rho_b}\left(\frac{\epsilon}{\rho_b}\right)\right]_{\rho_0} = 0 = \left[\frac{\partial}{\partial\rho_b}\left(\frac{\epsilon}{\rho_b}\right)\right]_{\rho_0} = \left[\frac{\partial\epsilon}{\partial\rho_b}\Big/\rho_b - \frac{\epsilon}{\rho_b^2}\right]_{\rho_0}$$

$$\Rightarrow \quad \left(\frac{\partial\epsilon}{\partial\rho_b}\right)_{\rho_0} = \left(\frac{\epsilon}{\rho_b}\right)_{\rho_0} = \left(\frac{B}{A}\right)_0 + m.$$

- Simplifying the derivative,

$$\begin{aligned}\frac{\partial\epsilon}{\partial\rho_b} &= \frac{\partial\langle E\rangle_{k_f}}{\partial\rho_b} + c_\omega^2\rho_b, \text{ with, } \langle E\rangle_{k_f} = \frac{2}{\pi^2}\int_0^{k_f} p^2\,dp\sqrt{p^2+m^{*2}} \;\&\; \rho_b = \frac{2k_f^3}{3\pi^2}. \\ &= \frac{\partial\langle E\rangle_{k_f}}{\partial k_f}\frac{\partial k_f}{\partial\rho_b} + c_\omega^2\rho_b\end{aligned}$$

Determination of c_ω

$\sigma - \omega - \rho$ model

- So we write,

$$\left[\frac{d}{d\rho_b}\left(\frac{\epsilon}{\rho_b}\right)\right]_{\rho_0} = 0 = \left[\frac{\partial}{\partial\rho_b}\left(\frac{\epsilon}{\rho_b}\right)\right]_{\rho_0} = \left[\frac{\partial\epsilon}{\partial\rho_b}\Big/\rho_b - \frac{\epsilon}{\rho_b^2}\right]_{\rho_0}$$

$$\Rightarrow \quad \left(\frac{\partial\epsilon}{\partial\rho_b}\right)_{\rho_0} = \left(\frac{\epsilon}{\rho_b}\right)_{\rho_0} = \left(\frac{B}{A}\right)_0 + m.$$

- Simplifying the derivative,

$$\begin{aligned}\frac{\partial\epsilon}{\partial\rho_b} &= \frac{\partial\langle E\rangle_{k_f}}{\partial\rho_b} + c_\omega^2\rho_b, \text{ with, } \langle E\rangle_{k_f} = \frac{2}{\pi^2}\int_0^{k_f} p^2\,dp\sqrt{p^2+m^{*2}} \;\&\; \rho_b = \frac{2k_f^3}{3\pi^2}. \\ &= \frac{\partial\langle E\rangle_{k_f}}{\partial k_f}\frac{\partial k_f}{\partial\rho_b} + c_\omega^2\rho_b = \sqrt{k_f^2+m^{*2}} + c_\omega^2\rho_b.\end{aligned}$$

Determination of c_ω

$\sigma - \omega - \rho$ model

- So we write,

$$\left[\frac{d}{d\rho_b}\left(\frac{\epsilon}{\rho_b}\right)\right]_{\rho_0} = 0 = \left[\frac{\partial}{\partial\rho_b}\left(\frac{\epsilon}{\rho_b}\right)\right]_{\rho_0} = \left[\frac{\partial\epsilon}{\partial\rho_b}\Big/\rho_b - \frac{\epsilon}{\rho_b^2}\right]_{\rho_0}$$

$$\Rightarrow \quad \left(\frac{\partial\epsilon}{\partial\rho_b}\right)_{\rho_0} = \left(\frac{\epsilon}{\rho_b}\right)_{\rho_0} = \left(\frac{B}{A}\right)_0 + m.$$

- Simplifying the derivative,

$$\begin{aligned}\frac{\partial\epsilon}{\partial\rho_b} &= \frac{\partial\langle E\rangle_{k_f}}{\partial\rho_b} + c_\omega^2\rho_b, \text{ with, } \langle E\rangle_{k_f} = \frac{2}{\pi^2}\int_0^{k_f} p^2\,dp\sqrt{p^2+m^{*2}} \;\&\; \rho_b = \frac{2k_f^3}{3\pi^2}. \\ &= \frac{\partial\langle E\rangle_{k_f}}{\partial k_f}\frac{\partial k_f}{\partial\rho_b} + c_\omega^2\rho_b = \sqrt{k_f^2+m^{*2}} + c_\omega^2\rho_b.\end{aligned}$$

- So we have

$$\left(\frac{B}{A}\right)_0 + m = \sqrt{k_{f_0}^2 + m_0^{*2}} + c_\omega^2\rho_0$$

Determination of c_ω

$\sigma - \omega - \rho$ model

- So we write,

$$\left[\frac{d}{d\rho_b}\left(\frac{\epsilon}{\rho_b}\right)\right]_{\rho_0} = 0 = \left[\frac{\partial}{\partial\rho_b}\left(\frac{\epsilon}{\rho_b}\right)\right]_{\rho_0} = \left[\frac{\partial\epsilon}{\partial\rho_b}\Big/\rho_b - \frac{\epsilon}{\rho_b^2}\right]_{\rho_0}$$

$$\Rightarrow \quad \left(\frac{\partial\epsilon}{\partial\rho_b}\right)_{\rho_0} = \left(\frac{\epsilon}{\rho_b}\right)_{\rho_0} = \left(\frac{B}{A}\right)_0 + m.$$

- Simplifying the derivative,

$$\begin{aligned}\frac{\partial\epsilon}{\partial\rho_b} &= \frac{\partial\langle E\rangle_{k_f}}{\partial\rho_b} + c_\omega^2\rho_b, \text{ with, } \langle E\rangle_{k_f} = \frac{2}{\pi^2}\int_0^{k_f} p^2\,dp\sqrt{p^2+m^{*2}} \;\&\; \rho_b = \frac{2k_f^3}{3\pi^2}. \\ &= \frac{\partial\langle E\rangle_{k_f}}{\partial k_f}\frac{\partial k_f}{\partial\rho_b} + c_\omega^2\rho_b = \sqrt{k_f^2+m^{*2}} + c_\omega^2\rho_b.\end{aligned}$$

- So we have

$$\left(\frac{B}{A}\right)_0 + m = \sqrt{k_{f_0}^2 + m_0^{*2}} + c_\omega^2\rho_0 \Rightarrow \boxed{c_\omega^2 = \frac{1}{\rho_0}\left[\left(\frac{B}{A}\right)_0 + m - \sqrt{k_{f_0}^2 + m_0^{*2}}\right]}$$

More on $V(\Phi)$ and derivatives

At saturation!!

More on $V(\Phi)$ and derivatives

At saturation!!

- The Dirac Eff. mass at ρ_0 : $m^* = m - g_\sigma \sigma = m - \Phi$

More on $V(\Phi)$ and derivatives

At saturation!!

- The Dirac Eff. mass at ρ_0 : $m^* = m - g_\sigma \sigma = m - \Phi \Rightarrow \Phi_0 = m - m_0^*$.

More on $V(\Phi)$ and derivatives

At saturation!!

- The Dirac Eff. mass at ρ_0 : $m^* = m - g_\sigma \sigma = m - \Phi \Rightarrow \Phi_0 = m - m_0^*$.

$$\text{Now, } V(\Phi_0) = \frac{1}{2}\frac{\Phi_0^2}{c_\sigma^2} + \frac{1}{3}\overline{g_2}\Phi_0^3 + \frac{1}{4}\overline{g_3}\Phi_0^4$$

More on $V(\Phi)$ and derivatives

At saturation!!

- The Dirac Eff. mass at ρ_0 : $m^* = m - g_\sigma \sigma = m - \Phi \Rightarrow \Phi_0 = m - m_0^*$.

$$\text{Now, } V(\Phi_0) = \frac{1}{2}\frac{\Phi_0^2}{c_\sigma^2} + \frac{1}{3}\overline{g_2}\Phi_0^3 + \frac{1}{4}\overline{g_3}\Phi_0^4$$

$$V'(\Phi_0) = \frac{\Phi_0}{c_\sigma^2} + \overline{g_2}\Phi_0^2 + \overline{g_3}\Phi_0^3$$

More on $V(\Phi)$ and derivatives

At saturation!!

- The Dirac Eff. mass at ρ_0 : $m^* = m - g_\sigma \sigma = m - \Phi \Rightarrow \Phi_0 = m - m_0^*$.

$$\begin{aligned}
\text{Now, } V(\Phi_0) &= \frac{1}{2}\frac{\Phi_0^2}{c_\sigma^2} + \frac{1}{3}\overline{g_2}\Phi_0^3 + \frac{1}{4}\overline{g_3}\Phi_0^4 \\
V'(\Phi_0) &= \frac{\Phi_0}{c_\sigma^2} + \overline{g_2}\Phi_0^2 + \overline{g_3}\Phi_0^3 \\
V''(\Phi_0) &= \frac{1}{c_\sigma^2} + 2\overline{g_2}\Phi_0 + 3\overline{g_3}\Phi_0^2
\end{aligned}$$

More on $V(\Phi)$ and derivatives

At saturation!!

- The Dirac Eff. mass at ρ_0 : $m^* = m - g_\sigma \sigma = m - \Phi \Rightarrow \Phi_0 = m - m_0^*$.

$$
\begin{aligned}
\text{Now, } V(\Phi_0) &= \frac{1}{2}\frac{\Phi_0^2}{c_\sigma^2} + \frac{1}{3}\overline{g_2}\Phi_0^3 + \frac{1}{4}\overline{g_3}\Phi_0^4 \\
V'(\Phi_0) &= \frac{\Phi_0}{c_\sigma^2} + \overline{g_2}\Phi_0^2 + \overline{g_3}\Phi_0^3 \\
V''(\Phi_0) &= \frac{1}{c_\sigma^2} + 2\overline{g_2}\Phi_0 + 3\overline{g_3}\Phi_0^2 \\
\Rightarrow \frac{1}{c_\sigma^2} &= \frac{2}{\Phi_0^2}\left[6\,V(\Phi_0) - 3\Phi_0\,V'(\Phi_0) + \frac{1}{2}\Phi_0^2\,V''(\Phi_0)\right] \\
\overline{g_2} &= \frac{3}{\Phi_0^3}\left[-8\,V(\Phi_0) + 5\Phi_0\,V'(\Phi_0) - \Phi_0^2\,V''(\Phi_0)\right] \\
\overline{g_3} &= \frac{4}{\Phi_0^4}\left[3\,V(\Phi_0) - 2\Phi_0\,V'(\Phi_0) + \frac{1}{2}\Phi_0^2\,V''(\Phi_0)\right]
\end{aligned}
$$

More on $V(\Phi)$ and derivatives

At saturation!!

- The Dirac Eff. mass at ρ_0 : $m^* = m - g_\sigma \sigma = m - \Phi \Rightarrow \Phi_0 = m - m_0^*$.

$$
\begin{aligned}
\text{Now, } V(\Phi_0) &= \frac{1}{2}\frac{\Phi_0^2}{c_\sigma^2} + \frac{1}{3}\overline{g_2}\Phi_0^3 + \frac{1}{4}\overline{g_3}\Phi_0^4 \\
V'(\Phi_0) &= \frac{\Phi_0}{c_\sigma^2} + \overline{g_2}\Phi_0^2 + \overline{g_3}\Phi_0^3 \\
V''(\Phi_0) &= \frac{1}{c_\sigma^2} + 2\overline{g_2}\Phi_0 + 3\overline{g_3}\Phi_0^2 \\
\Rightarrow \frac{1}{c_\sigma^2} &= \frac{2}{\Phi_0^2}\left[6\,V(\Phi_0) - 3\Phi_0\,V'(\Phi_0) + \frac{1}{2}\Phi_0^2\,V''(\Phi_0)\right] \\
\overline{g_2} &= \frac{3}{\Phi_0^3}\left[-8\,V(\Phi_0) + 5\Phi_0\,V'(\Phi_0) - \Phi_0^2\,V''(\Phi_0)\right] \\
\overline{g_3} &= \frac{4}{\Phi_0^4}\left[3\,V(\Phi_0) - 2\Phi_0\,V'(\Phi_0) + \frac{1}{2}\Phi_0^2\,V''(\Phi_0)\right]
\end{aligned}
$$

$$\text{So, } \epsilon_0 = \left[(B/A)_0 \quad + \quad m\right]\rho_0$$

More on $V(\Phi)$ and derivatives

At saturation!!

- The Dirac Eff. mass at ρ_0 : $m^* = m - g_\sigma \sigma = m - \Phi \Rightarrow \Phi_0 = m - m_0^*$.

$$
\begin{aligned}
\text{Now, } V(\Phi_0) &= \frac{1}{2}\frac{\Phi_0^2}{c_\sigma^2} + \frac{1}{3}\overline{g_2}\Phi_0^3 + \frac{1}{4}\overline{g_3}\Phi_0^4 \\
V'(\Phi_0) &= \frac{\Phi_0}{c_\sigma^2} + \overline{g_2}\Phi_0^2 + \overline{g_3}\Phi_0^3 \\
V''(\Phi_0) &= \frac{1}{c_\sigma^2} + 2\overline{g_2}\Phi_0 + 3\overline{g_3}\Phi_0^2 \\
\Rightarrow \frac{1}{c_\sigma^2} &= \frac{2}{\Phi_0^2}\left[6\,V(\Phi_0) - 3\Phi_0\,V'(\Phi_0) + \frac{1}{2}\Phi_0^2\,V''(\Phi_0)\right] \\
\overline{g_2} &= \frac{3}{\Phi_0^3}\left[-8\,V(\Phi_0) + 5\Phi_0\,V'(\Phi_0) - \Phi_0^2\,V''(\Phi_0)\right] \\
\overline{g_3} &= \frac{4}{\Phi_0^4}\left[3\,V(\Phi_0) - 2\Phi_0\,V'(\Phi_0) + \frac{1}{2}\Phi_0^2\,V''(\Phi_0)\right]
\end{aligned}
$$

$$
\text{So, } \epsilon_0 = \left[(B/A)_0 \quad + \quad m\right]\rho_0 = \langle E\rangle_{k_{f_0}} + V(\Phi_0) + \frac{1}{2}c_\omega^2\rho_0^2
$$

More on $V(\Phi)$ and derivatives

At saturation!!

- The Dirac Eff. mass at ρ_0 : $m^* = m - g_\sigma \sigma = m - \Phi \Rightarrow \Phi_0 = m - m_0^*$.

$$
\begin{aligned}
\text{Now, } V(\Phi_0) &= \frac{1}{2}\frac{\Phi_0^2}{c_\sigma^2} + \frac{1}{3}\overline{g_2}\Phi_0^3 + \frac{1}{4}\overline{g_3}\Phi_0^4 \\
V'(\Phi_0) &= \frac{\Phi_0}{c_\sigma^2} + \overline{g_2}\Phi_0^2 + \overline{g_3}\Phi_0^3 \\
V''(\Phi_0) &= \frac{1}{c_\sigma^2} + 2\overline{g_2}\Phi_0 + 3\overline{g_3}\Phi_0^2 \\
\Rightarrow \frac{1}{c_\sigma^2} &= \frac{2}{\Phi_0^2}\left[6\,V(\Phi_0) - 3\Phi_0\,V'(\Phi_0) + \frac{1}{2}\Phi_0^2\,V''(\Phi_0)\right] \\
\overline{g_2} &= \frac{3}{\Phi_0^3}\left[-8\,V(\Phi_0) + 5\Phi_0\,V'(\Phi_0) - \Phi_0^2\,V''(\Phi_0)\right] \\
\overline{g_3} &= \frac{4}{\Phi_0^4}\left[3\,V(\Phi_0) - 2\Phi_0\,V'(\Phi_0) + \frac{1}{2}\Phi_0^2\,V''(\Phi_0)\right]
\end{aligned}
$$

$$
\text{So, } \epsilon_0 = \left[(B/A)_0 \quad + \quad m\right]\rho_0 = \langle E\rangle_{k_{f_0}} + V(\Phi_0) + \frac{1}{2}c_\omega^2\rho_0^2
$$

So we have $\boxed{V(\Phi_0) = \left[(B/A)_0 + m\right]\rho_0 - \langle E\rangle_{k_{f_0}} - \frac{1}{2}c_\omega^2\rho_0^2}$

More on $V(\Phi)$ and derivatives

At saturation!!

- The Dirac Eff. mass at ρ_0 : $m^* = m - g_\sigma \sigma = m - \Phi \Rightarrow \Phi_0 = m - m_0^*$.

$$
\begin{aligned}
\text{Now, } V(\Phi_0) &= \frac{1}{2}\frac{\Phi_0^2}{c_\sigma^2} + \frac{1}{3}\overline{g_2}\Phi_0^3 + \frac{1}{4}\overline{g_3}\Phi_0^4 \\
V'(\Phi_0) &= \frac{\Phi_0}{c_\sigma^2} + \overline{g_2}\Phi_0^2 + \overline{g_3}\Phi_0^3 \\
V''(\Phi_0) &= \frac{1}{c_\sigma^2} + 2\overline{g_2}\Phi_0 + 3\overline{g_3}\Phi_0^2 \\
\Rightarrow \frac{1}{c_\sigma^2} &= \frac{2}{\Phi_0^2}\left[6\,V(\Phi_0) - 3\Phi_0\,V'(\Phi_0) + \frac{1}{2}\Phi_0^2\,V''(\Phi_0)\right] \\
\overline{g_2} &= \frac{3}{\Phi_0^3}\left[-8\,V(\Phi_0) + 5\Phi_0\,V'(\Phi_0) - \Phi_0^2\,V''(\Phi_0)\right] \\
\overline{g_3} &= \frac{4}{\Phi_0^4}\left[3\,V(\Phi_0) - 2\Phi_0\,V'(\Phi_0) + \frac{1}{2}\Phi_0^2\,V''(\Phi_0)\right]
\end{aligned}
$$

$$
\text{So, } \epsilon_0 = \left[(B/A)_0 \quad + \quad m\right]\rho_0 = \langle E\rangle_{k_{f_0}} + V(\Phi_0) + \frac{1}{2}c_\omega^2\rho_0^2
$$

So we have $\boxed{V(\Phi_0) = \left[(B/A)_0 + m\right]\rho_0 - \langle E\rangle_{k_{f_0}} - \frac{1}{2}c_\omega^2\rho_0^2}$ & $\boxed{V'(\Phi_0) = \rho_{s,0}}$.

More on $V(\Phi)$ and derivatives

Incompressibility at saturation!

More on $V(\Phi)$ and derivatives

Incompressibility at saturation!

- From the definition of pressure

$$\text{We know, } \mathtt{p}(\rho_b) = \rho_b^2 \frac{d\left(\epsilon/\rho_b\right)}{d\rho_b}$$

More on $V(\Phi)$ and derivatives

Incompressibility at saturation!

- From the definition of pressure

$$\text{We know, } \mathtt{p}(\rho_b) = \rho_b^2 \frac{d\left(\epsilon/\rho_b\right)}{d\rho_b} = \rho_b \left(\frac{d\epsilon}{d\rho_b}\right) - \epsilon$$

More on $V(\Phi)$ and derivatives

Incompressibility at saturation!

- From the definition of pressure

$$\text{We know, } \mathtt{p}(\rho_b) = \rho_b^2 \frac{d\left(\epsilon/\rho_b\right)}{d\rho_b} = \rho_b \left(\frac{d\epsilon}{d\rho_b}\right) - \epsilon$$

$$\text{Now, } K_0 = 9\rho_0^2 \left(\frac{d^2\left(\epsilon/\rho_b\right)}{d\rho_b^2}\right)_{\rho_0}$$

More on $V(\Phi)$ and derivatives

Incompressibility at saturation!

- From the definition of pressure

$$\text{We know, } \mathtt{p}(\rho_b) = \rho_b^2 \frac{d\left(\epsilon/\rho_b\right)}{d\rho_b} = \rho_b \left(\frac{d\epsilon}{d\rho_b}\right) - \epsilon$$

$$\text{Now, } K_0 = 9\rho_0^2 \left(\frac{d^2(\epsilon/\rho_b)}{d\rho_b^2}\right)_{\rho_0} \Rightarrow \frac{d^2(\epsilon/\rho_b)}{d\rho_b^2} = \frac{d}{d\rho_b}\left[\frac{1}{\rho_b}\left(\frac{d\epsilon}{d\rho_b}\right) - \frac{\epsilon}{\rho_b^2}\right]$$

More on $V(\Phi)$ and derivatives

Incompressibility at saturation!

- From the definition of pressure

$$\text{We know, } \mathtt{p}(\rho_b) = \rho_b^2 \frac{d\left(\epsilon/\rho_b\right)}{d\rho_b} = \rho_b \left(\frac{d\epsilon}{d\rho_b}\right) - \epsilon$$

$$\text{Now, } K_0 = 9\rho_0^2 \left(\frac{d^2(\epsilon/\rho_b)}{d\rho_b^2}\right)_{\rho_0} \Rightarrow \frac{d^2(\epsilon/\rho_b)}{d\rho_b^2} = \frac{d}{d\rho_b}\left[\frac{1}{\rho_b}\left(\frac{d\epsilon}{d\rho_b}\right) - \frac{\epsilon}{\rho_b^2}\right]$$

$$\Rightarrow \quad \frac{d^2(\epsilon/\rho_b)}{d\rho_b^2} = \frac{1}{\rho_b}\frac{d^2\epsilon}{d\rho_b^2} - \frac{2}{\rho_b^3}\left[\rho_b \frac{d\epsilon}{d\rho_b} - \epsilon\right]$$

More on $V(\Phi)$ and derivatives

Incompressibility at saturation!

- From the definition of pressure

$$\text{We know, } \mathtt{p}(\rho_b) = \rho_b^2 \frac{d\left(\epsilon/\rho_b\right)}{d\rho_b} = \rho_b \left(\frac{d\epsilon}{d\rho_b}\right) - \epsilon$$

$$\text{Now, } K_0 = 9\rho_0^2 \left(\frac{d^2(\epsilon/\rho_b)}{d\rho_b^2}\right)_{\rho_0} \Rightarrow \frac{d^2(\epsilon/\rho_b)}{d\rho_b^2} = \frac{d}{d\rho_b}\left[\frac{1}{\rho_b}\left(\frac{d\epsilon}{d\rho_b}\right) - \frac{\epsilon}{\rho_b^2}\right]$$

$$\Rightarrow \quad \frac{d^2(\epsilon/\rho_b)}{d\rho_b^2} = \frac{1}{\rho_b}\frac{d^2\epsilon}{d\rho_b^2} - \frac{2}{\rho_b^3}\left[\rho_b \frac{d\epsilon}{d\rho_b} - \epsilon\right] = \frac{1}{\rho_b}\left[\frac{d^2\epsilon}{d\rho_b^2} - \frac{2\mathtt{p}}{\rho_b^2}\right]$$

More on $V(\Phi)$ and derivatives

Incompressibility at saturation!

- From the definition of pressure

$$\text{We know, } \mathtt{p}(\rho_b) = \rho_b^2 \frac{d\left(\epsilon/\rho_b\right)}{d\rho_b} = \rho_b \left(\frac{d\epsilon}{d\rho_b}\right) - \epsilon$$

$$\text{Now, } K_0 = 9\rho_0^2 \left(\frac{d^2(\epsilon/\rho_b)}{d\rho_b^2}\right)_{\rho_0} \Rightarrow \frac{d^2(\epsilon/\rho_b)}{d\rho_b^2} = \frac{d}{d\rho_b}\left[\frac{1}{\rho_b}\left(\frac{d\epsilon}{d\rho_b}\right) - \frac{\epsilon}{\rho_b^2}\right]$$

$$\Rightarrow \quad \frac{d^2(\epsilon/\rho_b)}{d\rho_b^2} = \frac{1}{\rho_b}\frac{d^2\epsilon}{d\rho_b^2} - \frac{2}{\rho_b^3}\left[\rho_b \frac{d\epsilon}{d\rho_b} - \epsilon\right] = \frac{1}{\rho_b}\left[\frac{d^2\epsilon}{d\rho_b^2} - \frac{2\mathtt{p}}{\rho_b^2}\right]$$

$$\Rightarrow \quad K_0 = 9\rho_0 \left[\frac{d^2\epsilon}{d\rho_b^2} - \frac{2\mathtt{p}}{\rho_b^2}\right]_{\rho_0}$$

More on $V(\Phi)$ and derivatives

Incompressibility at saturation!

- From the definition of pressure

$$\text{We know, } \mathtt{p}(\rho_b) = \rho_b^2 \frac{d\left(\epsilon/\rho_b\right)}{d\rho_b} = \rho_b \left(\frac{d\epsilon}{d\rho_b}\right) - \epsilon$$

$$\text{Now, } K_0 = 9\rho_0^2 \left(\frac{d^2(\epsilon/\rho_b)}{d\rho_b^2}\right)_{\rho_0} \Rightarrow \frac{d^2(\epsilon/\rho_b)}{d\rho_b^2} = \frac{d}{d\rho_b}\left[\frac{1}{\rho_b}\left(\frac{d\epsilon}{d\rho_b}\right) - \frac{\epsilon}{\rho_b^2}\right]$$

$$\Rightarrow \quad \frac{d^2(\epsilon/\rho_b)}{d\rho_b^2} = \frac{1}{\rho_b}\frac{d^2\epsilon}{d\rho_b^2} - \frac{2}{\rho_b^3}\left[\rho_b \frac{d\epsilon}{d\rho_b} - \epsilon\right] = \frac{1}{\rho_b}\left[\frac{d^2\epsilon}{d\rho_b^2} - \frac{2\mathtt{p}}{\rho_b^2}\right]$$

$$\Rightarrow \quad K_0 = 9\rho_0 \left[\frac{d^2\epsilon}{d\rho_b^2} - \frac{2\mathtt{p}}{\rho_b^2}\right]_{\rho_0} = 9\rho_0 \left(\frac{d^2\epsilon}{d\rho_b^2}\right)_{\rho_0}, \text{ as, } \mathtt{p}(\rho_0) = 0$$

More on $V(\Phi)$ and derivatives

Incompressibility at saturation!

- From the definition of pressure

$$\text{We know, } \mathtt{p}(\rho_b) = \rho_b^2 \frac{d\left(\epsilon/\rho_b\right)}{d\rho_b} = \rho_b \left(\frac{d\epsilon}{d\rho_b}\right) - \epsilon$$

$$\text{Now, } K_0 = 9\rho_0^2 \left(\frac{d^2(\epsilon/\rho_b)}{d\rho_b^2}\right)_{\rho_0} \Rightarrow \frac{d^2(\epsilon/\rho_b)}{d\rho_b^2} = \frac{d}{d\rho_b}\left[\frac{1}{\rho_b}\left(\frac{d\epsilon}{d\rho_b}\right) - \frac{\epsilon}{\rho_b^2}\right]$$

$$\Rightarrow \quad \frac{d^2(\epsilon/\rho_b)}{d\rho_b^2} = \frac{1}{\rho_b}\frac{d^2\epsilon}{d\rho_b^2} - \frac{2}{\rho_b^3}\left[\rho_b \frac{d\epsilon}{d\rho_b} - \epsilon\right] = \frac{1}{\rho_b}\left[\frac{d^2\epsilon}{d\rho_b^2} - \frac{2\mathtt{p}}{\rho_b^2}\right]$$

$$\Rightarrow \quad K_0 = 9\rho_0 \left[\frac{d^2\epsilon}{d\rho_b^2} - \frac{2\mathtt{p}}{\rho_b^2}\right]_{\rho_0} = 9\rho_0 \left(\frac{d^2\epsilon}{d\rho_b^2}\right)_{\rho_0}, \text{ as, } \mathtt{p}(\rho_0) = 0$$

$$= 9\rho_0 \left[\frac{d^2}{d\rho_b^2}\left(\langle E\rangle_{k_f} + V(\Phi) + \frac{1}{2}c_\omega^2 \rho_b^2\right)\right]_{\rho_0}$$

More on $V(\Phi)$ and derivatives

Incompressibility at saturation!

- From the definition of pressure

$$\text{We know, } \mathtt{p}(\rho_b) = \rho_b^2 \frac{d\left(\epsilon/\rho_b\right)}{d\rho_b} = \rho_b \left(\frac{d\epsilon}{d\rho_b}\right) - \epsilon$$

$$\text{Now, } K_0 = 9\rho_0^2 \left(\frac{d^2(\epsilon/\rho_b)}{d\rho_b^2}\right)_{\rho_0} \Rightarrow \frac{d^2(\epsilon/\rho_b)}{d\rho_b^2} = \frac{d}{d\rho_b}\left[\frac{1}{\rho_b}\left(\frac{d\epsilon}{d\rho_b}\right) - \frac{\epsilon}{\rho_b^2}\right]$$

$$\Rightarrow \quad \frac{d^2(\epsilon/\rho_b)}{d\rho_b^2} = \frac{1}{\rho_b}\frac{d^2\epsilon}{d\rho_b^2} - \frac{2}{\rho_b^3}\left[\rho_b \frac{d\epsilon}{d\rho_b} - \epsilon\right] = \frac{1}{\rho_b}\left[\frac{d^2\epsilon}{d\rho_b^2} - \frac{2\mathtt{p}}{\rho_b^2}\right]$$

$$\Rightarrow \quad K_0 = 9\rho_0 \left[\frac{d^2\epsilon}{d\rho_b^2} - \frac{2\mathtt{p}}{\rho_b^2}\right]_{\rho_0} = 9\rho_0 \left(\frac{d^2\epsilon}{d\rho_b^2}\right)_{\rho_0}, \text{ as, } \mathtt{p}(\rho_0) = 0$$

$$= 9\rho_0 \left[\frac{d^2}{d\rho_b^2}\left(\langle E\rangle_{k_f} + V(\Phi) + \frac{1}{2}c_\omega^2 \rho_b^2\right)\right]_{\rho_0}$$

$$= 9\rho_0 \left[\frac{d}{d\rho_b}\left(\sqrt{k_f^2 + m^{*2}} + c_\omega^2 \rho_b\right)\right]_{\rho_0}, \text{ as, } \frac{\partial\epsilon}{\partial\Phi} = 0$$

More on $V(\Phi)$ and derivatives

Incompressibility at saturation!

- From the definition of pressure

$$\text{We know, } \mathtt{p}(\rho_b) = \rho_b^2 \frac{d\left(\epsilon/\rho_b\right)}{d\rho_b} = \rho_b \left(\frac{d\epsilon}{d\rho_b}\right) - \epsilon$$

$$\text{Now, } K_0 = 9\rho_0^2 \left(\frac{d^2(\epsilon/\rho_b)}{d\rho_b^2}\right)_{\rho_0} \Rightarrow \frac{d^2(\epsilon/\rho_b)}{d\rho_b^2} = \frac{d}{d\rho_b}\left[\frac{1}{\rho_b}\left(\frac{d\epsilon}{d\rho_b}\right) - \frac{\epsilon}{\rho_b^2}\right]$$

$$\Rightarrow \quad \frac{d^2(\epsilon/\rho_b)}{d\rho_b^2} = \frac{1}{\rho_b}\frac{d^2\epsilon}{d\rho_b^2} - \frac{2}{\rho_b^3}\left[\rho_b \frac{d\epsilon}{d\rho_b} - \epsilon\right] = \frac{1}{\rho_b}\left[\frac{d^2\epsilon}{d\rho_b^2} - \frac{2\mathtt{p}}{\rho_b^2}\right]$$

$$\Rightarrow \quad K_0 = 9\rho_0 \left[\frac{d^2\epsilon}{d\rho_b^2} - \frac{2\mathtt{p}}{\rho_b^2}\right]_{\rho_0} = 9\rho_0 \left(\frac{d^2\epsilon}{d\rho_b^2}\right)_{\rho_0}, \text{ as, } \mathtt{p}(\rho_0) = 0$$

$$= 9\rho_0 \left[\frac{d^2}{d\rho_b^2}\left(\langle E\rangle_{k_f} + V(\Phi) + \frac{1}{2}c_\omega^2 \rho_b^2\right)\right]_{\rho_0}$$

$$= 9\rho_0 \left[\frac{d}{d\rho_b}\left(\sqrt{k_f^2 + m^{*2}} + c_\omega^2 \rho_b\right)\right]_{\rho_0}, \text{ as, } \frac{\partial\epsilon}{\partial\Phi} = 0$$

$$= 9\rho_0 \left[\frac{\partial}{\partial k_f}\left(\sqrt{k_f^2 + m^{*2}}\right)\frac{\partial k_f}{\partial \rho_b} + \frac{\partial}{\partial\Phi}\left(\sqrt{k_f^2 + m^{*2}}\right)\frac{\partial\Phi}{\partial\rho_b} + c_\omega^2\right]_{\rho_0}$$

More on $V(\Phi)$ and derivatives

Incompressibility at saturation!

More on $V(\Phi)$ and derivatives

Incompressibility at saturation!

- So we have

$$K_0 = 9\rho_0 \left[\frac{k_f}{\sqrt{k_f^2 + m^{*2}}} \frac{\pi^2}{2k_f^2} - \frac{m^*}{\sqrt{k_f^2 + m^{*2}}} \frac{\partial \Phi}{\partial \rho_b} + c_\omega^2 \right]_{\rho_0}$$

More on $V(\Phi)$ and derivatives

Incompressibility at saturation!

- So we have

$$K_0 = 9\rho_0 \left[\frac{k_f}{\sqrt{k_f^2 + m^{*2}}} \frac{\pi^2}{2k_f^2} - \frac{m^*}{\sqrt{k_f^2 + m^{*2}}} \frac{\partial \Phi}{\partial \rho_b} + c_\omega^2 \right]_{\rho_0}$$

Now, $V'(\Phi) = \rho_s$

More on $V(\Phi)$ and derivatives

Incompressibility at saturation!

- So we have

$$K_0 = 9\rho_0 \left[\frac{k_f}{\sqrt{k_f^2 + m^{*2}}} \frac{\pi^2}{2k_f^2} - \frac{m^*}{\sqrt{k_f^2 + m^{*2}}} \frac{\partial \Phi}{\partial \rho_b} + c_\omega^2 \right]_{\rho_0}$$

$$\text{Now,} \quad V'(\Phi) = \rho_s \Rightarrow \frac{d}{d\rho_b} V'(\Phi) = \frac{d\rho_s}{d\rho_b}$$

More on $V(\Phi)$ and derivatives

Incompressibility at saturation!

- So we have

$$K_0 = 9\rho_0 \left[\frac{k_f}{\sqrt{k_f^2 + m^{*2}}} \frac{\pi^2}{2k_f^2} - \frac{m^*}{\sqrt{k_f^2 + m^{*2}}} \frac{\partial \Phi}{\partial \rho_b} + c_\omega^2 \right]_{\rho_0}$$

$$\text{Now,} \quad V'(\Phi) = \rho_s \Rightarrow \frac{d}{d\rho_b} V'(\Phi) = \frac{d\rho_s}{d\rho_b} \Rightarrow V''(\Phi) \frac{\partial \Phi}{\partial \rho_b} = \frac{\partial \rho_s}{\partial \rho_b} + \frac{\partial \rho_s}{\partial \Phi} \frac{\partial \Phi}{\partial \rho_b}$$

More on $V(\Phi)$ and derivatives

Incompressibility at saturation!

- So we have

$$K_0 = 9\rho_0 \left[\frac{k_f}{\sqrt{k_f^2 + m^{*2}}} \frac{\pi^2}{2k_f^2} - \frac{m^*}{\sqrt{k_f^2 + m^{*2}}} \frac{\partial \Phi}{\partial \rho_b} + c_\omega^2 \right]_{\rho_0}$$

$$\text{Now,} \quad V'(\Phi) = \rho_s \Rightarrow \frac{d}{d\rho_b} V'(\Phi) = \frac{d\rho_s}{d\rho_b} \Rightarrow V''(\Phi) \frac{\partial \Phi}{\partial \rho_b} = \frac{\partial \rho_s}{\partial \rho_b} + \frac{\partial \rho_s}{\partial \Phi} \frac{\partial \Phi}{\partial \rho_b}$$

$$\Rightarrow \quad V''(\Phi) \frac{\partial \Phi}{\partial \rho_b} = \frac{m^*}{\sqrt{k_f^2 + m^{*2}}} + \rho_s' \frac{\partial \Phi}{\partial \rho_b}, \text{ as, } \rho_s = \frac{2}{\pi^2} \int_0^{k_f} p^2 \, dp \frac{m - g_\sigma \sigma}{\sqrt{p^2 + (m - g_\sigma \sigma)^2}}$$

More on $V(\Phi)$ and derivatives

Incompressibility at saturation!

- So we have

$$K_0 = 9\rho_0 \left[\frac{k_f}{\sqrt{k_f^2 + m^{*2}}} \frac{\pi^2}{2k_f^2} - \frac{m^*}{\sqrt{k_f^2 + m^{*2}}} \frac{\partial \Phi}{\partial \rho_b} + c_\omega^2 \right]_{\rho_0}$$

$$\text{Now,} \quad V'(\Phi) = \rho_s \Rightarrow \frac{d}{d\rho_b} V'(\Phi) = \frac{d\rho_s}{d\rho_b} \Rightarrow V''(\Phi) \frac{\partial \Phi}{\partial \rho_b} = \frac{\partial \rho_s}{\partial \rho_b} + \frac{\partial \rho_s}{\partial \Phi} \frac{\partial \Phi}{\partial \rho_b}$$

$$\Rightarrow \quad V''(\Phi) \frac{\partial \Phi}{\partial \rho_b} = \frac{m^*}{\sqrt{k_f^2 + m^{*2}}} + \rho_s' \frac{\partial \Phi}{\partial \rho_b}, \text{ as, } \rho_s = \frac{2}{\pi^2} \int_0^{k_f} p^2 \, dp \frac{m - g_\sigma \sigma}{\sqrt{p^2 + (m - g_\sigma \sigma)^2}}$$

$$\Rightarrow \quad \frac{\partial \Phi}{\partial \rho_b} = \left(m^* \Big/ \sqrt{k_f^2 + m^{*2}} \right) \Big/ \left(V''(\Phi) - \rho_s' \right)$$

More on $V(\Phi)$ and derivatives

Incompressibility at saturation!

- So we have

$$K_0 = 9\rho_0 \left[\frac{k_f}{\sqrt{k_f^2 + m^{*2}}} \frac{\pi^2}{2k_f^2} - \frac{m^*}{\sqrt{k_f^2 + m^{*2}}} \frac{\partial \Phi}{\partial \rho_b} + c_\omega^2 \right]_{\rho_0}$$

$$\text{Now,} \quad V'(\Phi) = \rho_s \Rightarrow \frac{d}{d\rho_b} V'(\Phi) = \frac{d\rho_s}{d\rho_b} \Rightarrow V''(\Phi) \frac{\partial \Phi}{\partial \rho_b} = \frac{\partial \rho_s}{\partial \rho_b} + \frac{\partial \rho_s}{\partial \Phi} \frac{\partial \Phi}{\partial \rho_b}$$

$$\Rightarrow \quad V''(\Phi) \frac{\partial \Phi}{\partial \rho_b} = \frac{m^*}{\sqrt{k_f^2 + m^{*2}}} + \rho_s' \frac{\partial \Phi}{\partial \rho_b}, \text{ as, } \rho_s = \frac{2}{\pi^2} \int_0^{k_f} p^2 \, dp \frac{m - g_\sigma \sigma}{\sqrt{p^2 + (m - g_\sigma \sigma)^2}}$$

$$\Rightarrow \quad \frac{\partial \Phi}{\partial \rho_b} = \left(m^* \Big/ \sqrt{k_f^2 + m^{*2}} \right) \Big/ \left(V''(\Phi) - \rho_s' \right)$$

$$\Rightarrow \quad \frac{K_0}{9\rho_0} = \left[\frac{\pi^2}{2k_f \sqrt{k_f^2 + m^{*2}}} - \frac{\left(m^* / \sqrt{k_f^2 + m^{*2}} \right)^2}{\left(V''(\Phi) - \rho_s' \right)} + c_\omega^2 \right]_{\rho_0}$$

More on $V(\Phi)$ and derivatives

Incompressibility at saturation!

- So we have

$$K_0 = 9\rho_0 \left[\frac{k_f}{\sqrt{k_f^2 + m^{*2}}} \frac{\pi^2}{2k_f^2} - \frac{m^*}{\sqrt{k_f^2 + m^{*2}}} \frac{\partial \Phi}{\partial \rho_b} + c_\omega^2 \right]_{\rho_0}$$

$$\text{Now,} \quad V'(\Phi) = \rho_s \Rightarrow \frac{d}{d\rho_b} V'(\Phi) = \frac{d\rho_s}{d\rho_b} \Rightarrow V''(\Phi) \frac{\partial \Phi}{\partial \rho_b} = \frac{\partial \rho_s}{\partial \rho_b} + \frac{\partial \rho_s}{\partial \Phi} \frac{\partial \Phi}{\partial \rho_b}$$

$$\Rightarrow \quad V''(\Phi) \frac{\partial \Phi}{\partial \rho_b} = \frac{m^*}{\sqrt{k_f^2 + m^{*2}}} + \rho_s' \frac{\partial \Phi}{\partial \rho_b}, \text{ as, } \rho_s = \frac{2}{\pi^2} \int_0^{k_f} p^2 \, dp \frac{m - g_\sigma \sigma}{\sqrt{p^2 + (m - g_\sigma \sigma)^2}}$$

$$\Rightarrow \quad \frac{\partial \Phi}{\partial \rho_b} = \left(m^* \Big/ \sqrt{k_f^2 + m^{*2}} \right) \Big/ \left(V''(\Phi) - \rho_s' \right)$$

$$\Rightarrow \quad \frac{K_0}{9\rho_0} = \left[\frac{\pi^2}{2k_f \sqrt{k_f^2 + m^{*2}}} - \frac{\left(m^* / \sqrt{k_f^2 + m^{*2}} \right)^2}{\left(V''(\Phi) - \rho_s' \right)} + c_\omega^2 \right]_{\rho_0}$$

$$\Rightarrow \quad \frac{\left(m_0^* / \mu_0 \right)^2}{\left(V''(\Phi_0) - \rho_{s,0}' \right)} = \frac{\pi^2}{2k_{f_0} \mu_0} + c_\omega^2 - \frac{K_0}{9\rho_0}, \text{ with, } \mu_0 = \sqrt{k_{f_0}^2 + m_0^{*2}}$$

Parameters of SNM

$\sigma - \omega - \boldsymbol{\rho}$ model

Parameters of SNM

$\sigma - \omega - \rho$ model

- So we have now all the $V(\Phi)$ and their derivatives as

$$\boxed{V''(\Phi_0) = \rho'_{s,0} + \frac{\left(m_0^*/\mu_0\right)^2}{\frac{\pi^2}{2k_{f_0}\mu_0} + c_\omega^2 - \frac{K_0}{9\rho_0}}}$$

Parameters of SNM

$\sigma - \omega - \rho$ model

- So we have now all the $V(\Phi)$ and their derivatives as

$$\boxed{V''(\Phi_0) = \rho'_{s,0} + \frac{\left(m_0^*/\mu_0\right)^2}{\frac{\pi^2}{2k_{f_0}\mu_0} + c_\omega^2 - \frac{K_0}{9\rho_0}}}$$

$$\boxed{V(\Phi_0) = \left[(B/A)_0 + m\right]\rho_0 - \langle E\rangle_{k_{f_0}} - \frac{1}{2}c_\omega^2\rho_0^2}$$

Parameters of SNM

$\sigma - \omega - \rho$ model

- So we have now all the $V(\Phi)$ and their derivatives as

$$\boxed{V''(\Phi_0) = \rho'_{s,0} + \frac{\left(m_0^*/\mu_0\right)^2}{\frac{\pi^2}{2k_{f_0}\mu_0} + c_\omega^2 - \frac{K_0}{9\rho_0}}}$$

$$\boxed{V(\Phi_0) = \left[(B/A)_0 + m\right]\rho_0 - \langle E\rangle_{k_{f_0}} - \frac{1}{2}c_\omega^2\rho_0^2}$$

$$\boxed{V'(\Phi_0) = \rho_{s,0}}$$

Parameters of SNM

$\sigma - \omega - \rho$ model

- So we have now all the $V(\Phi)$ and their derivatives as

$$\boxed{V''(\Phi_0) = \rho'_{s,0} + \frac{\left(m_0^*/\mu_0\right)^2}{\frac{\pi^2}{2k_{f_0}\mu_0} + c_\omega^2 - \frac{K_0}{9\rho_0}}}$$

$$\boxed{V(\Phi_0) = \left[(B/A)_0 + m\right]\rho_0 - \langle E\rangle_{k_{f_0}} - \frac{1}{2}c_\omega^2\rho_0^2}$$

$$\boxed{V'(\Phi_0) = \rho_{s,0}}$$

One can find $c_\sigma, \overline{g_2}$ and $\overline{g_3}$ with this.

Parameters of SNM

$\sigma - \omega - \rho$ model

- So we have now all the $V(\Phi)$ and their derivatives as

$$\boxed{V''(\Phi_0) = \rho'_{s,0} + \frac{(m_0^*/\mu_0)^2}{\frac{\pi^2}{2k_{f_0}\mu_0} + c_\omega^2 - \frac{K_0}{9\rho_0}}}$$

$$\boxed{V(\Phi_0) = \left[(B/A)_0 + m\right]\rho_0 - \langle E\rangle_{k_{f_0}} - \frac{1}{2}c_\omega^2\rho_0^2}$$

$$\boxed{V'(\Phi_0) = \rho_{s,0}}$$

One can find $c_\sigma, \overline{g_2}$ and $\overline{g_3}$ with this. We also have

$$\boxed{c_\omega^2 = \frac{1}{\rho_0}\left[\left(\frac{B}{A}\right)_0 + m - \sqrt{k_{f_0}^2 + m_0^{*2}}\right]}$$

Parameters of SNM

$\sigma - \omega - \rho$ model

- So we have now all the $V(\Phi)$ and their derivatives as

$$\boxed{V''(\Phi_0) = \rho'_{s,0} + \frac{(m_0^*/\mu_0)^2}{\frac{\pi^2}{2k_{f_0}\mu_0} + c_\omega^2 - \frac{K_0}{9\rho_0}}}$$

$$\boxed{V(\Phi_0) = \left[(B/A)_0 + m\right]\rho_0 - \langle E\rangle_{k_{f_0}} - \frac{1}{2}c_\omega^2\rho_0^2}$$

$$\boxed{V'(\Phi_0) = \rho_{s,0}}$$

One can find $c_\sigma, \overline{g_2}$ and $\overline{g_3}$ with this. We also have

$$\boxed{c_\omega^2 = \frac{1}{\rho_0}\left[\left(\frac{B}{A}\right)_0 + m - \sqrt{k_{f_0}^2 + m_0^{*2}}\right]}$$

Using the constraints from SNM at saturation one can find these coupling constants.

Symmetry energy

$\sigma - \omega - \boldsymbol{\rho}$ model

Symmetry energy

$\sigma - \omega - \boldsymbol{\rho}$ model

- With the same philosophy as σ, ω mesons we define,

$$c_{\boldsymbol{\rho}} = \frac{\mathrm{g}_{\boldsymbol{\rho}}}{2}\frac{1}{m_{\boldsymbol{\rho}}}, \; R_3 = \frac{\mathrm{g}_{\boldsymbol{\rho}}}{2}\rho_3^0$$

Symmetry energy

$\sigma - \omega - \boldsymbol{\rho}$ model

- With the same philosophy as σ, ω mesons we define,

$$c_{\boldsymbol{\rho}} = \frac{g_{\boldsymbol{\rho}}}{2}\frac{1}{m_{\boldsymbol{\rho}}}, \; R_3 = \frac{g_{\boldsymbol{\rho}}}{2}\rho_3^0$$

- So from field equation we have,

$$m_{\boldsymbol{\rho}}^2 \rho_3^0 = \frac{g_{\boldsymbol{\rho}}}{2}(\rho_n - \rho_p)$$

Symmetry energy

$\sigma - \omega - \boldsymbol{\rho}$ model

- With the same philosophy as σ, ω mesons we define,

$$c_{\boldsymbol{\rho}} = \frac{g_{\boldsymbol{\rho}}}{2}\frac{1}{m_{\boldsymbol{\rho}}}, \; R_3 = \frac{g_{\boldsymbol{\rho}}}{2}\rho_3^0$$

- So from field equation we have,

$$m_{\boldsymbol{\rho}}^2 \rho_3^0 = \frac{g_{\boldsymbol{\rho}}}{2}(\rho_n - \rho_p) \Rightarrow R_3 = c_{\boldsymbol{\rho}}^2(\rho_n - \rho_p)$$

Symmetry energy

$\sigma - \omega - \boldsymbol{\rho}$ model

- With the same philosophy as σ, ω mesons we define,

$$c_{\boldsymbol{\rho}} = \frac{g_{\boldsymbol{\rho}}}{2}\frac{1}{m_{\boldsymbol{\rho}}}, \; R_3 = \frac{g_{\boldsymbol{\rho}}}{2}\rho_3^0$$

- So from field equation we have,

$$m_{\boldsymbol{\rho}}^2 \rho_3^0 = \frac{g_{\boldsymbol{\rho}}}{2}(\rho_n - \rho_p) \Rightarrow R_3 = c_{\boldsymbol{\rho}}^2(\rho_n - \rho_p)$$

- The contribution to the energy apart from interaction is given by

$$\frac{1}{2} m_{\boldsymbol{\rho}}^2 \rho_{3,0}^2 = \frac{1}{2}\frac{R_3^2}{c_{\boldsymbol{\rho}}^2}$$

Symmetry energy

$\sigma - \omega - \boldsymbol{\rho}$ model

- With the same philosophy as σ, ω mesons we define,

$$c_{\boldsymbol{\rho}} = \frac{g_{\boldsymbol{\rho}}}{2}\frac{1}{m_{\boldsymbol{\rho}}}, \; R_3 = \frac{g_{\boldsymbol{\rho}}}{2}\rho_3^0$$

- So from field equation we have,

$$m_{\boldsymbol{\rho}}^2 \rho_3^0 = \frac{g_{\boldsymbol{\rho}}}{2}(\rho_n - \rho_p) \Rightarrow R_3 = c_{\boldsymbol{\rho}}^2(\rho_n - \rho_p)$$

- The contribution to the energy apart from interaction is given by

$$\frac{1}{2}m_{\boldsymbol{\rho}}^2 \rho_{3,0}^2 = \frac{1}{2}\frac{R_3^2}{c_{\boldsymbol{\rho}}^2} = \frac{1}{2}c_{\boldsymbol{\rho}}^2(\rho_n - \rho_p)^2$$

Symmetry energy

$\sigma - \omega - \boldsymbol{\rho}$ model

- With the same philosophy as σ, ω mesons we define,

$$c_{\boldsymbol{\rho}} = \frac{g_{\boldsymbol{\rho}}}{2}\frac{1}{m_{\boldsymbol{\rho}}}, \;\; R_3 = \frac{g_{\boldsymbol{\rho}}}{2}\rho_3^0$$

- So from field equation we have,

$$m_{\boldsymbol{\rho}}^2 \rho_3^0 = \frac{g_{\boldsymbol{\rho}}}{2}(\rho_n - \rho_p) \Rightarrow R_3 = c_{\boldsymbol{\rho}}^2(\rho_n - \rho_p)$$

- The contribution to the energy apart from interaction is given by

$$\frac{1}{2}m_{\boldsymbol{\rho}}^2\rho_{3,0}^2 = \frac{1}{2}\frac{R_3^2}{c_{\boldsymbol{\rho}}^2} = \frac{1}{2}c_{\boldsymbol{\rho}}^2(\rho_n - \rho_p)^2 = \frac{1}{2}c_{\boldsymbol{\rho}}^2\rho_b^2\delta^2, \;\; \text{as}, \;\; \delta = \frac{\rho_n - \rho_p}{\rho_b}$$

Symmetry energy

$\sigma - \omega - \boldsymbol{\rho}$ model

- With the same philosophy as σ, ω mesons we define,

$$c_{\boldsymbol{\rho}} = \frac{g_{\boldsymbol{\rho}}}{2}\frac{1}{m_{\boldsymbol{\rho}}}, \;\; R_3 = \frac{g_{\boldsymbol{\rho}}}{2}\rho_3^0$$

- So from field equation we have,

$$m_{\boldsymbol{\rho}}^2 \rho_3^0 = \frac{g_{\boldsymbol{\rho}}}{2}(\rho_n - \rho_p) \Rightarrow R_3 = c_{\boldsymbol{\rho}}^2(\rho_n - \rho_p)$$

- The contribution to the energy apart from interaction is given by

$$\frac{1}{2}m_{\boldsymbol{\rho}}^2\rho_{3,0}^2 = \frac{1}{2}\frac{R_3^2}{c_{\boldsymbol{\rho}}^2} = \frac{1}{2}c_{\boldsymbol{\rho}}^2(\rho_n - \rho_p)^2 = \frac{1}{2}c_{\boldsymbol{\rho}}^2\rho_b^2\delta^2, \;\; \text{as}, \;\; \delta = \frac{\rho_n - \rho_p}{\rho_b}$$

- The symmetry energy is defined as

$$e_{\text{sym}}(\rho_b) = \frac{1}{2}\left(\frac{d^2\left(\epsilon/\rho_b\right)}{d\delta^2}\right)_{\delta=0}$$

Symmetry energy

$\sigma - \omega - \boldsymbol{\rho}$ model

- With the same philosophy as σ, ω mesons we define,

$$c_{\boldsymbol{\rho}} = \frac{g_{\boldsymbol{\rho}}}{2}\frac{1}{m_{\boldsymbol{\rho}}}, \;\; R_3 = \frac{g_{\boldsymbol{\rho}}}{2}\rho_3^0$$

- So from field equation we have,

$$m_{\boldsymbol{\rho}}^2 \rho_3^0 = \frac{g_{\boldsymbol{\rho}}}{2}(\rho_n - \rho_p) \Rightarrow R_3 = c_{\boldsymbol{\rho}}^2(\rho_n - \rho_p)$$

- The contribution to the energy apart from interaction is given by

$$\frac{1}{2}m_{\boldsymbol{\rho}}^2\rho_{3,0}^2 = \frac{1}{2}\frac{R_3^2}{c_{\boldsymbol{\rho}}^2} = \frac{1}{2}c_{\boldsymbol{\rho}}^2(\rho_n - \rho_p)^2 = \frac{1}{2}c_{\boldsymbol{\rho}}^2\rho_b^2\delta^2, \;\; \text{as}, \;\; \delta = \frac{\rho_n - \rho_p}{\rho_b}$$

- The symmetry energy is defined as

$$e_{\text{sym}}(\rho_b) = \frac{1}{2}\left(\frac{d^2\left(\epsilon/\rho_b\right)}{d\delta^2}\right)_{\delta=0} = \frac{1}{2\rho_b}\left(\frac{d^2\epsilon}{d\delta^2}\right)_{\delta=0}$$

Symmetry energy

$\sigma - \omega - \boldsymbol{\rho}$ model

Symmetry energy

$\sigma - \omega - \boldsymbol{\rho}$ model

- The energy density is written as

$$\epsilon = \sum_{q=n,p} \langle E \rangle_{k_{f_q}} + V(\Phi) + \frac{1}{2} c_\omega^2 \rho_b^2 + \frac{1}{2} c_{\boldsymbol{\rho}}^2 \rho_b^2 \delta^2, \text{ with, } k_{f_q} = k_f (1 \pm \delta)^{1/3}$$

Symmetry energy

$\sigma - \omega - \boldsymbol{\rho}$ model

- The energy density is written as

$$\epsilon = \sum_{q=n,p} \langle E \rangle_{k_{f_q}} + V(\Phi) + \frac{1}{2} c_\omega^2 \rho_b^2 + \frac{1}{2} c_{\boldsymbol{\rho}}^2 \rho_b^2 \delta^2, \text{ with, } k_{f_q} = k_f (1 \pm \delta)^{1/3}$$

$$\Rightarrow \quad \frac{d^2 \epsilon}{d\delta^2} = \frac{d^2}{d\delta^2} \sum_{q=n,p} \langle E \rangle_{k_{f_q}} + c_{\boldsymbol{\rho}}^2 \rho_b^2, \text{ with, } \langle E \rangle_{k_{f_q}} = \frac{1}{\pi^2} \int_0^{k_{f_q}} p^2 \, dp \sqrt{p^2 + m^{*\,2}}$$

Symmetry energy

$\sigma - \omega - \boldsymbol{\rho}$ model

- The energy density is written as

$$\epsilon = \sum_{q=n,p} \langle E \rangle_{k_{f_q}} + V(\Phi) + \frac{1}{2} c_\omega^2 \rho_b^2 + \frac{1}{2} c_{\boldsymbol{\rho}}^2 \rho_b^2 \delta^2, \text{ with, } k_{f_q} = k_f (1 \pm \delta)^{1/3}$$

$$\Rightarrow \quad \frac{d^2 \epsilon}{d\delta^2} = \frac{d^2}{d\delta^2} \sum_{q=n,p} \langle E \rangle_{k_{f_q}} + c_{\boldsymbol{\rho}}^2 \rho_b^2, \text{ with, } \langle E \rangle_{k_{f_q}} = \frac{1}{\pi^2} \int_0^{k_{f_q}} p^2 \, dp \sqrt{p^2 + m^{*2}}$$

- So the derivatives are evaluated as,

$$\frac{d\langle E \rangle_{k_{f_q}}}{d\delta} = \frac{\partial \langle E \rangle_{k_{f_q}}}{\partial k_{f_q}} \frac{\partial k_{f_q}}{\partial \delta}$$

Symmetry energy

$\sigma - \omega - \boldsymbol{\rho}$ model

- The energy density is written as

$$\epsilon = \sum_{q=n,p} \langle E \rangle_{k_{f_q}} + V(\Phi) + \frac{1}{2} c_\omega^2 \rho_b^2 + \frac{1}{2} c_{\boldsymbol{\rho}}^2 \rho_b^2 \delta^2, \text{ with, } k_{f_q} = k_f (1 \pm \delta)^{1/3}$$

$$\Rightarrow \quad \frac{d^2 \epsilon}{d\delta^2} = \frac{d^2}{d\delta^2} \sum_{q=n,p} \langle E \rangle_{k_{f_q}} + c_{\boldsymbol{\rho}}^2 \rho_b^2, \text{ with, } \langle E \rangle_{k_{f_q}} = \frac{1}{\pi^2} \int_0^{k_{f_q}} p^2 \, dp \sqrt{p^2 + m^{*2}}$$

- So the derivatives are evaluated as,

$$\frac{d\langle E \rangle_{k_{f_q}}}{d\delta} = \frac{\partial \langle E \rangle_{k_{f_q}}}{\partial k_{f_q}} \frac{\partial k_{f_q}}{\partial \delta} \Rightarrow \frac{d^2 \langle E \rangle_{k_{f_q}}}{d\delta^2} = \frac{\partial^2 \langle E \rangle_{k_{f_q}}}{\partial k_{f_q}^2} \left(\frac{\partial k_{f_q}}{\partial \delta} \right)^2 + \frac{\partial \langle E \rangle_{k_{f_q}}}{\partial k_{f_q}} \frac{\partial^2 k_{f_q}}{\partial \delta^2}$$

Symmetry energy

$\sigma - \omega - \boldsymbol{\rho}$ model

- The energy density is written as

$$\epsilon = \sum_{q=n,p} \langle E \rangle_{k_{f_q}} + V(\Phi) + \frac{1}{2} c_\omega^2 \rho_b^2 + \frac{1}{2} c_{\boldsymbol{\rho}}^2 \rho_b^2 \delta^2, \text{ with, } k_{f_q} = k_f (1 \pm \delta)^{1/3}$$

$$\Rightarrow \quad \frac{d^2 \epsilon}{d\delta^2} = \frac{d^2}{d\delta^2} \sum_{q=n,p} \langle E \rangle_{k_{f_q}} + c_{\boldsymbol{\rho}}^2 \rho_b^2, \text{ with, } \langle E \rangle_{k_{f_q}} = \frac{1}{\pi^2} \int_0^{k_{f_q}} p^2 \, dp \sqrt{p^2 + m^{*2}}$$

- So the derivatives are evaluated as,

$$\frac{d\langle E \rangle_{k_{f_q}}}{d\delta} = \frac{\partial \langle E \rangle_{k_{f_q}}}{\partial k_{f_q}} \frac{\partial k_{f_q}}{\partial \delta} \Rightarrow \frac{d^2 \langle E \rangle_{k_{f_q}}}{d\delta^2} = \frac{\partial^2 \langle E \rangle_{k_{f_q}}}{\partial k_{f_q}^2} \left(\frac{\partial k_{f_q}}{\partial \delta} \right)^2 + \frac{\partial \langle E \rangle_{k_{f_q}}}{\partial k_{f_q}} \frac{\partial^2 k_{f_q}}{\partial \delta^2}$$

Where, $\dfrac{\partial k_{f_q}}{\partial \delta} = \pm \dfrac{k_f}{3} (1 \pm \delta)^{-2/3}$

Symmetry energy

$\sigma - \omega - \boldsymbol{\rho}$ model

- The energy density is written as

$$\epsilon = \sum_{q=n,p} \langle E \rangle_{k_{f_q}} + V(\Phi) + \frac{1}{2} c_\omega^2 \rho_b^2 + \frac{1}{2} c_{\boldsymbol{\rho}}^2 \rho_b^2 \delta^2, \text{ with, } k_{f_q} = k_f (1 \pm \delta)^{1/3}$$

$$\Rightarrow \quad \frac{d^2 \epsilon}{d\delta^2} = \frac{d^2}{d\delta^2} \sum_{q=n,p} \langle E \rangle_{k_{f_q}} + c_{\boldsymbol{\rho}}^2 \rho_b^2, \text{ with, } \langle E \rangle_{k_{f_q}} = \frac{1}{\pi^2} \int_0^{k_{f_q}} p^2 \, dp \sqrt{p^2 + m^{*2}}$$

- So the derivatives are evaluated as,

$$\frac{d\langle E \rangle_{k_{f_q}}}{d\delta} = \frac{\partial \langle E \rangle_{k_{f_q}}}{\partial k_{f_q}} \frac{\partial k_{f_q}}{\partial \delta} \Rightarrow \frac{d^2 \langle E \rangle_{k_{f_q}}}{d\delta^2} = \frac{\partial^2 \langle E \rangle_{k_{f_q}}}{\partial k_{f_q}^2} \left(\frac{\partial k_{f_q}}{\partial \delta} \right)^2 + \frac{\partial \langle E \rangle_{k_{f_q}}}{\partial k_{f_q}} \frac{\partial^2 k_{f_q}}{\partial \delta^2}$$

$$\text{Where, } \frac{\partial k_{f_q}}{\partial \delta} = \pm \frac{k_f}{3} (1 \pm \delta)^{-2/3} \quad \Rightarrow \quad \frac{\partial^2 k_{f_q}}{\partial \delta^2} = -\frac{2 k_f}{9} (1 \pm \delta)^{-5/3}$$

Symmetry energy

$\sigma - \omega - \boldsymbol{\rho}$ model

- The energy density is written as

$$\epsilon = \sum_{q=n,p} \langle E \rangle_{k_{f_q}} + V(\Phi) + \frac{1}{2} c_\omega^2 \rho_b^2 + \frac{1}{2} c_{\boldsymbol{\rho}}^2 \rho_b^2 \delta^2, \text{ with, } k_{f_q} = k_f (1 \pm \delta)^{1/3}$$

$$\Rightarrow \quad \frac{d^2 \epsilon}{d\delta^2} = \frac{d^2}{d\delta^2} \sum_{q=n,p} \langle E \rangle_{k_{f_q}} + c_{\boldsymbol{\rho}}^2 \rho_b^2, \text{ with, } \langle E \rangle_{k_{f_q}} = \frac{1}{\pi^2} \int_0^{k_{f_q}} p^2 \, dp \sqrt{p^2 + m^{*2}}$$

- So the derivatives are evaluated as,

$$\frac{d\langle E \rangle_{k_{f_q}}}{d\delta} = \frac{\partial \langle E \rangle_{k_{f_q}}}{\partial k_{f_q}} \frac{\partial k_{f_q}}{\partial \delta} \Rightarrow \frac{d^2 \langle E \rangle_{k_{f_q}}}{d\delta^2} = \frac{\partial^2 \langle E \rangle_{k_{f_q}}}{\partial k_{f_q}^2} \left(\frac{\partial k_{f_q}}{\partial \delta} \right)^2 + \frac{\partial \langle E \rangle_{k_{f_q}}}{\partial k_{f_q}} \frac{\partial^2 k_{f_q}}{\partial \delta^2}$$

$$\text{Where, } \frac{\partial k_{f_q}}{\partial \delta} = \pm \frac{k_f}{3} (1 \pm \delta)^{-2/3} \quad \Rightarrow \quad \frac{\partial^2 k_{f_q}}{\partial \delta^2} = -\frac{2 k_f}{9} (1 \pm \delta)^{-5/3}$$

$$\text{And, } \frac{\partial \langle E \rangle_{k_{f_q}}}{\partial k_{f_q}} = \frac{1}{\pi^2} k_{f_q}^2 \sqrt{k_{f_q}^2 + m^{*2}}$$

Symmetry energy

$\sigma - \omega - \boldsymbol{\rho}$ model

- The energy density is written as

$$\epsilon = \sum_{q=n,p} \langle E \rangle_{k_{f_q}} + V(\Phi) + \frac{1}{2} c_\omega^2 \rho_b^2 + \frac{1}{2} c_{\boldsymbol{\rho}}^2 \rho_b^2 \delta^2, \text{ with, } k_{f_q} = k_f (1 \pm \delta)^{1/3}$$

$$\Rightarrow \quad \frac{d^2 \epsilon}{d\delta^2} = \frac{d^2}{d\delta^2} \sum_{q=n,p} \langle E \rangle_{k_{f_q}} + c_{\boldsymbol{\rho}}^2 \rho_b^2, \text{ with, } \langle E \rangle_{k_{f_q}} = \frac{1}{\pi^2} \int_0^{k_{f_q}} p^2 \, dp \sqrt{p^2 + m^{*2}}$$

- So the derivatives are evaluated as,

$$\frac{d\langle E \rangle_{k_{f_q}}}{d\delta} = \frac{\partial \langle E \rangle_{k_{f_q}}}{\partial k_{f_q}} \frac{\partial k_{f_q}}{\partial \delta} \Rightarrow \frac{d^2 \langle E \rangle_{k_{f_q}}}{d\delta^2} = \frac{\partial^2 \langle E \rangle_{k_{f_q}}}{\partial k_{f_q}^2} \left(\frac{\partial k_{f_q}}{\partial \delta} \right)^2 + \frac{\partial \langle E \rangle_{k_{f_q}}}{\partial k_{f_q}} \frac{\partial^2 k_{f_q}}{\partial \delta^2}$$

$$\text{Where, } \frac{\partial k_{f_q}}{\partial \delta} = \pm \frac{k_f}{3} (1 \pm \delta)^{-2/3} \quad \Rightarrow \quad \frac{\partial^2 k_{f_q}}{\partial \delta^2} = -\frac{2 k_f}{9} (1 \pm \delta)^{-5/3}$$

$$\text{And, } \frac{\partial \langle E \rangle_{k_{f_q}}}{\partial k_{f_q}} = \frac{1}{\pi^2} k_{f_q}^2 \sqrt{k_{f_q}^2 + m^{*2}}$$

$$\Rightarrow \frac{\partial^2 \langle E \rangle_{k_{f_q}}}{\partial k_{f_q}^2} = \frac{1}{\pi^2} \left(2 k_{f_q} \sqrt{k_{f_q}^2 + m^{*2}} + k_{f_q}^3 \Big/ \sqrt{k_{f_q}^2 + m^{*2}} \right)$$

Symmetry energy

$\sigma - \omega - \boldsymbol{\rho}$ model

Symmetry energy

$\sigma - \omega - \boldsymbol{\rho}$ model

- Combining the results we can write

$$\frac{d^2 \langle E \rangle_{k_{f_q}}}{d\delta^2} = \frac{1}{\pi^2}\left(2k_{f_q}\sqrt{k_{f_q}^2 + m^{*2}} + k_{f_q}^3 \Big/ \sqrt{k_{f_q}^2 + m^{*2}}\right)\frac{k_f^2}{9}(1 \pm \delta)^{-4/3} - \frac{1}{\pi^2}k_{f_q}^2\sqrt{k_{f_q}^2 + m^{*2}}\;\frac{2k_f}{9}(1 \pm \delta)^{-5/3}$$

Symmetry energy

$\sigma - \omega - \rho$ model

- Combining the results we can write

$$\begin{aligned}
\frac{d^2 \langle E \rangle_{k_{f_q}}}{d\delta^2} &= \frac{1}{\pi^2}\left(2k_{f_q}\sqrt{k_{f_q}^2 + m^{*2}} + k_{f_q}^3 \Big/ \sqrt{k_{f_q}^2 + m^{*2}}\right)\frac{k_f^2}{9}(1 \pm \delta)^{-4/3} \\
&\quad - \frac{1}{\pi^2} k_{f_q}^2 \sqrt{k_{f_q}^2 + m^{*2}}\ \frac{2k_f}{9}(1 \pm \delta)^{-5/3} \\
&= \frac{1}{\pi^2}\left(k_{f_q}^3 \Big/ \sqrt{k_{f_q}^2 + m^{*2}}\right)\frac{k_f^2}{9}(1 \pm \delta)^{-4/3}
\end{aligned}$$

Symmetry energy

$\sigma - \omega - \rho$ model

- Combining the results we can write

$$\begin{aligned}
\frac{d^2 \langle E \rangle_{k_{f_q}}}{d\delta^2} &= \frac{1}{\pi^2}\left(2k_{f_q}\sqrt{k_{f_q}^2 + m^{*2}} + k_{f_q}^3 \Big/ \sqrt{k_{f_q}^2 + m^{*2}}\right)\frac{k_f^2}{9}(1 \pm \delta)^{-4/3} \\
&\quad - \frac{1}{\pi^2} k_{f_q}^2 \sqrt{k_{f_q}^2 + m^{*2}}\ \frac{2k_f}{9}(1 \pm \delta)^{-5/3} \\
&= \frac{1}{\pi^2}\left(k_{f_q}^3 \Big/ \sqrt{k_{f_q}^2 + m^{*2}}\right)\frac{k_f^2}{9}(1 \pm \delta)^{-4/3} \\
\left.\frac{d^2 \langle E \rangle_{k_{f_q}}}{d\delta^2}\right|_{\delta=0} &= \frac{1}{9\pi^2}\left(k_f^5 \Big/ \sqrt{k_f^2 + m^{*2}}\right)
\end{aligned}$$

Symmetry energy

$\sigma - \omega - \rho$ model

- Combining the results we can write

$$\begin{aligned}
\frac{d^2 \langle E \rangle_{k_{f_q}}}{d\delta^2} &= \frac{1}{\pi^2}\left(2k_{f_q}\sqrt{k_{f_q}^2 + m^{*2}} + k_{f_q}^3 \Big/ \sqrt{k_{f_q}^2 + m^{*2}}\right)\frac{k_f^2}{9}(1 \pm \delta)^{-4/3} \\
&\quad - \frac{1}{\pi^2} k_{f_q}^2 \sqrt{k_{f_q}^2 + m^{*2}}\ \frac{2k_f}{9}(1 \pm \delta)^{-5/3} \\
&= \frac{1}{\pi^2}\left(k_{f_q}^3 \Big/ \sqrt{k_{f_q}^2 + m^{*2}}\right)\frac{k_f^2}{9}(1 \pm \delta)^{-4/3} \\
\left.\frac{d^2 \langle E \rangle_{k_{f_q}}}{d\delta^2}\right|_{\delta=0} &= \frac{1}{9\pi^2}\left(k_f^5 \Big/ \sqrt{k_f^2 + m^{*2}}\right) = \frac{\rho_b}{6}\left(k_f^2 \Big/ \sqrt{k_f^2 + m^{*2}}\right)
\end{aligned}$$

Symmetry energy

$\sigma - \omega - \boldsymbol{\rho}$ model

- Combining the results we can write

$$\begin{aligned}
\frac{d^2 \langle E \rangle_{k_{f_q}}}{d\delta^2} &= \frac{1}{\pi^2}\left(2k_{f_q}\sqrt{k_{f_q}^2 + m^{*2}} + k_{f_q}^3 \Big/ \sqrt{k_{f_q}^2 + m^{*2}}\right)\frac{k_f^2}{9}(1 \pm \delta)^{-4/3} \\
&\quad - \frac{1}{\pi^2} k_{f_q}^2 \sqrt{k_{f_q}^2 + m^{*2}}\ \frac{2k_f}{9}(1 \pm \delta)^{-5/3} \\
&= \frac{1}{\pi^2}\left(k_{f_q}^3 \Big/ \sqrt{k_{f_q}^2 + m^{*2}}\right)\frac{k_f^2}{9}(1 \pm \delta)^{-4/3} \\
\left.\frac{d^2 \langle E \rangle_{k_{f_q}}}{d\delta^2}\right|_{\delta=0} &= \frac{1}{9\pi^2}\left(k_f^5 \Big/ \sqrt{k_f^2 + m^{*2}}\right) = \frac{\rho_b}{6}\left(k_f^2 \Big/ \sqrt{k_f^2 + m^{*2}}\right)
\end{aligned}$$

- So finally we write symmetry energy as,

$$e_{\mathrm{sym}}(\rho_b) = \frac{1}{2\rho_b}\left(\frac{d^2}{d\delta^2}\sum_{q=n,p}\langle E \rangle_{k_{f_q}} + c_{\boldsymbol{\rho}}^2 \rho_b^2\right)_{\delta=0}$$

Symmetry energy

$\sigma - \omega - \boldsymbol{\rho}$ model

- Combining the results we can write

$$\begin{aligned}
\frac{d^2 \langle E \rangle_{k_{f_q}}}{d\delta^2} &= \frac{1}{\pi^2}\left(2k_{f_q}\sqrt{k_{f_q}^2 + m^{*2}} + k_{f_q}^3 \Big/ \sqrt{k_{f_q}^2 + m^{*2}}\right)\frac{k_f^2}{9}(1 \pm \delta)^{-4/3} \\
&\quad - \frac{1}{\pi^2} k_{f_q}^2 \sqrt{k_{f_q}^2 + m^{*2}}\ \frac{2k_f}{9}(1 \pm \delta)^{-5/3} \\
&= \frac{1}{\pi^2}\left(k_{f_q}^3 \Big/ \sqrt{k_{f_q}^2 + m^{*2}}\right)\frac{k_f^2}{9}(1 \pm \delta)^{-4/3} \\
\left.\frac{d^2 \langle E \rangle_{k_{f_q}}}{d\delta^2}\right|_{\delta=0} &= \frac{1}{9\pi^2}\left(k_f^5 \Big/ \sqrt{k_f^2 + m^{*2}}\right) = \frac{\rho_b}{6}\left(k_f^2 \Big/ \sqrt{k_f^2 + m^{*2}}\right)
\end{aligned}$$

- So finally we write symmetry energy as,

$$e_{\mathrm{sym}}(\rho_b) = \frac{1}{2\rho_b}\left(\frac{d^2}{d\delta^2}\sum_{q=n,p}\langle E \rangle_{k_{f_q}} + c_{\boldsymbol{\rho}}^2 \rho_b^2\right)_{\delta=0} = \frac{k_f^2}{6\sqrt{k_f^2 + m^{*2}}} + \left(\frac{g_{\boldsymbol{\rho}}^2}{4m_{\boldsymbol{\rho}}^2}\right)\frac{k_f^3}{3\pi^2}$$

Symmetry energy

$\sigma - \omega - \boldsymbol{\rho}$ model

- Combining the results we can write

$$\begin{aligned}
\frac{d^2 \langle E \rangle_{k_{f_q}}}{d\delta^2} &= \frac{1}{\pi^2}\left(2k_{f_q}\sqrt{k_{f_q}^2 + m^{*2}} + k_{f_q}^3 \Big/ \sqrt{k_{f_q}^2 + m^{*2}}\right)\frac{k_f^2}{9}(1 \pm \delta)^{-4/3} \\
&\quad - \frac{1}{\pi^2} k_{f_q}^2 \sqrt{k_{f_q}^2 + m^{*2}}\ \frac{2k_f}{9}(1 \pm \delta)^{-5/3} \\
&= \frac{1}{\pi^2}\left(k_{f_q}^3 \Big/ \sqrt{k_{f_q}^2 + m^{*2}}\right)\frac{k_f^2}{9}(1 \pm \delta)^{-4/3} \\
\left.\frac{d^2 \langle E \rangle_{k_{f_q}}}{d\delta^2}\right|_{\delta=0} &= \frac{1}{9\pi^2}\left(k_f^5 \Big/ \sqrt{k_f^2 + m^{*2}}\right) = \frac{\rho_b}{6}\left(k_f^2 \Big/ \sqrt{k_f^2 + m^{*2}}\right)
\end{aligned}$$

- So finally we write symmetry energy as,

$$e_{\mathrm{sym}}(\rho_b) = \frac{1}{2\rho_b}\left(\frac{d^2}{d\delta^2}\sum_{q=n,p}\langle E \rangle_{k_{f_q}} + c_{\boldsymbol{\rho}}^2 \rho_b^2\right)_{\delta=0} = \frac{k_f^2}{6\sqrt{k_f^2 + m^{*2}}} + \left(\frac{g_{\boldsymbol{\rho}}^2}{4m_{\boldsymbol{\rho}}^2}\right)\frac{k_f^3}{3\pi^2}$$

$$\Rightarrow e_{\mathrm{sym}}(\rho_0) = J$$

Symmetry energy

$\sigma-\omega-\boldsymbol{\rho}$ model

- Combining the results we can write

$$\begin{aligned}
\frac{d^2\langle E\rangle_{k_{f_q}}}{d\delta^2} &= \frac{1}{\pi^2}\left(2k_{f_q}\sqrt{k_{f_q}^2+m^{*2}}+k_{f_q}^3\Big/\sqrt{k_{f_q}^2+m^{*2}}\right)\frac{k_f^2}{9}(1\pm\delta)^{-4/3} \\
&\quad -\frac{1}{\pi^2}k_{f_q}^2\sqrt{k_{f_q}^2+m^{*2}}\;\frac{2k_f}{9}(1\pm\delta)^{-5/3} \\
&= \frac{1}{\pi^2}\left(k_{f_q}^3\Big/\sqrt{k_{f_q}^2+m^{*2}}\right)\frac{k_f^2}{9}(1\pm\delta)^{-4/3} \\
\left.\frac{d^2\langle E\rangle_{k_{f_q}}}{d\delta^2}\right|_{\delta=0} &= \frac{1}{9\pi^2}\left(k_f^5\Big/\sqrt{k_f^2+m^{*2}}\right) = \frac{\rho_b}{6}\left(k_f^2\Big/\sqrt{k_f^2+m^{*2}}\right)
\end{aligned}$$

- So finally we write symmetry energy as,

$$\begin{aligned}
e_{\rm sym}(\rho_b) &= \frac{1}{2\rho_b}\left(\frac{d^2}{d\delta^2}\sum_{q=n,p}\langle E\rangle_{k_{f_q}}+c_{\boldsymbol{\rho}}^2\rho_b^2\right)_{\delta=0} = \frac{k_f^2}{6\sqrt{k_f^2+m^{*2}}}+\left(\frac{g_{\boldsymbol{\rho}}^2}{4m_{\boldsymbol{\rho}}^2}\right)\frac{k_f^3}{3\pi^2} \\
\Rightarrow e_{\rm sym}(\rho_0) &= J \Rightarrow \boxed{\frac{g_{\boldsymbol{\rho}}}{2}=\left[\left(J-k_{f_0}^2\Big/6\sqrt{k_{f_0}^2+m_0^{*2}}\right)\frac{3\pi^2}{k_{f_0}^3}m_{\boldsymbol{\rho}}^2\right]^{1/2}}
\end{aligned}$$

Computer algorithm

$\sigma - \omega - \rho$ model

Computer algorithm

$\sigma - \omega - \boldsymbol{\rho}$ model

- Input as constraints:

$$\rho_0, (B/A)_0, K_0, m_0^*, J, m, m_\sigma, m_\omega \text{ and } m_{\boldsymbol{\rho}}$$

Computer algorithm

$\sigma - \omega - \boldsymbol{\rho}$ model

- Input as constraints:

$$\rho_0, (B/A)_0, K_0, m_0^*, J, m, m_\sigma, m_\omega \text{ and } m_{\boldsymbol{\rho}}$$

$$\Phi_0 = m - m_0^*,$$

Computer algorithm

$\sigma - \omega - \boldsymbol{\rho}$ model

- Input as constraints:

$$\rho_0, (B/A)_0, K_0, m_0^*, J, m, m_\sigma, m_\omega \text{ and } m_{\boldsymbol{\rho}}$$

$$\Phi_0 = m - m_0^*, \; k_{f_0} = \left(\frac{3\pi^2\rho_0}{2}\right)^{1/3},$$

Computer algorithm

$\sigma - \omega - \boldsymbol{\rho}$ model

- Input as constraints:

$$\rho_0, (B/A)_0, K_0, m_0^*, J, m, m_\sigma, m_\omega \text{ and } m_{\boldsymbol{\rho}}$$

$$\Phi_0 = m - m_0^*, \; k_{f_0} = \left(\frac{3\pi^2 \rho_0}{2}\right)^{1/3}, \; \mu_0 = \sqrt{k_{f_0}^2 + m_0^{*2}}$$

Computer algorithm

$\sigma - \omega - \boldsymbol{\rho}$ model

- Input as constraints:

$$\rho_0, (B/A)_0, K_0, m_0^*, J, m, m_\sigma, m_\omega \text{ and } m_{\boldsymbol{\rho}}$$

$$\Phi_0 = m - m_0^*, \; k_{f_0} = \left(\frac{3\pi^2\rho_0}{2}\right)^{1/3}, \; \mu_0 = \sqrt{k_{f_0}^2 + m_0^{*2}}$$

$$\langle E \rangle_{k_{f_0}} = \frac{2}{\pi^2}\int_0^{k_{f_0}} p^2\, dp \sqrt{p^2 + m_0^{*2}},$$

Computer algorithm

$\sigma - \omega - \boldsymbol{\rho}$ model

- Input as constraints:

$$\rho_0, (B/A)_0, K_0, m_0^*, J, m, m_\sigma, m_\omega \text{ and } m_{\boldsymbol{\rho}}$$

$$\Phi_0 = m - m_0^*, \; k_{f_0} = \left(\frac{3\pi^2\rho_0}{2}\right)^{1/3}, \; \mu_0 = \sqrt{k_{f_0}^2 + m_0^{*2}}$$

$$\langle E \rangle_{k_{f_0}} = \frac{2}{\pi^2}\int_0^{k_{f_0}} p^2\, dp \sqrt{p^2 + m_0^{*2}}, \; \rho_{s,0} = \frac{2}{\pi^2}\int_0^{k_{f_0}} p^2\, dp \frac{m_0^*}{\sqrt{p^2 + m_0^{*2}}}$$

Computer algorithm

$\sigma - \omega - \boldsymbol{\rho}$ model

- Input as constraints:

$$\rho_0, (B/A)_0, K_0, m_0^*, J, m, m_\sigma, m_\omega \text{ and } m_{\boldsymbol{\rho}}$$

$$\Phi_0 = m - m_0^*, \; k_{f_0} = \left(\frac{3\pi^2\rho_0}{2}\right)^{1/3}, \; \mu_0 = \sqrt{k_{f_0}^2 + m_0^{*2}}$$

$$\langle E \rangle_{k_{f_0}} = \frac{2}{\pi^2}\int_0^{k_{f_0}} p^2\, dp \sqrt{p^2 + m_0^{*2}}, \; \rho_{s,0} = \frac{2}{\pi^2}\int_0^{k_{f_0}} p^2\, dp \frac{m_0^*}{\sqrt{p^2 + m_0^{*2}}}$$

$$\rho'_{s,0} = \frac{2}{\pi^2}\int_0^{k_{f_0}} p^2\, dp \left[m_0^{*2} \Big/ \left(\sqrt{p^2 + m_0^{*2}}\right)^3 - \frac{1}{\sqrt{p^2 + m_0^{*2}}} \right]$$

Computer algorithm

$\sigma - \omega - \boldsymbol{\rho}$ model

- Input as constraints:

$$\rho_0, (B/A)_0, K_0, m_0^*, J, m, m_\sigma, m_\omega \text{ and } m_{\boldsymbol{\rho}}$$

$$\Phi_0 = m - m_0^*, \; k_{f_0} = \left(\frac{3\pi^2\rho_0}{2}\right)^{1/3}, \; \mu_0 = \sqrt{k_{f_0}^2 + m_0^{*2}}$$

$$\langle E \rangle_{k_{f_0}} = \frac{2}{\pi^2}\int_0^{k_{f_0}} p^2\, dp \sqrt{p^2 + m_0^{*2}}, \; \rho_{s,0} = \frac{2}{\pi^2}\int_0^{k_{f_0}} p^2\, dp \frac{m_0^*}{\sqrt{p^2 + m_0^{*2}}}$$

$$\rho'_{s,0} = \frac{2}{\pi^2}\int_0^{k_{f_0}} p^2\, dp \left[m_0^{*2} \Big/ \left(\sqrt{p^2 + m_0^{*2}}\right)^3 - \frac{1}{\sqrt{p^2 + m_0^{*2}}} \right]$$

$$c_\omega^2 = \frac{1}{\rho_0}\left[(B/A)_0 + m - \sqrt{k_{f_0}^2 + m_0^{*2}}\right],$$

Computer algorithm

$\sigma - \omega - \boldsymbol{\rho}$ model

- Input as constraints:

$$\rho_0, (B/A)_0, K_0, m_0^*, J, m, m_\sigma, m_\omega \text{ and } m_{\boldsymbol{\rho}}$$

$$\Phi_0 = m - m_0^*, \; k_{f_0} = \left(\frac{3\pi^2\rho_0}{2}\right)^{1/3}, \; \mu_0 = \sqrt{k_{f_0}^2 + m_0^{*2}}$$

$$\langle E \rangle_{k_{f_0}} = \frac{2}{\pi^2}\int_0^{k_{f_0}} p^2\, dp \sqrt{p^2 + m_0^{*2}}, \; \rho_{s,0} = \frac{2}{\pi^2}\int_0^{k_{f_0}} p^2\, dp \frac{m_0^*}{\sqrt{p^2 + m_0^{*2}}}$$

$$\rho'_{s,0} = \frac{2}{\pi^2}\int_0^{k_{f_0}} p^2\, dp \left[m_0^{*2} \Big/ \left(\sqrt{p^2 + m_0^{*2}}\right)^3 - \frac{1}{\sqrt{p^2 + m_0^{*2}}} \right]$$

$$c_\omega^2 = \frac{1}{\rho_0}\left[(B/A)_0 + m - \sqrt{k_{f_0}^2 + m_0^{*2}}\right],$$

$$V(\Phi_0) = \left[(B/A)_0 + m\right]\rho_0 - \langle E \rangle_{k_{f_0}} - \frac{1}{2}c_\omega^2\rho_0^2,$$

Computer algorithm

$\sigma - \omega - \boldsymbol{\rho}$ model

- Input as constraints:

$$\rho_0, (B/A)_0, K_0, m_0^*, J, m, m_\sigma, m_\omega \text{ and } m_{\boldsymbol{\rho}}$$

$$\Phi_0 = m - m_0^*, \; k_{f_0} = \left(\frac{3\pi^2\rho_0}{2}\right)^{1/3}, \; \mu_0 = \sqrt{k_{f_0}^2 + m_0^{*2}}$$

$$\langle E \rangle_{k_{f_0}} = \frac{2}{\pi^2}\int_0^{k_{f_0}} p^2\, dp \sqrt{p^2 + m_0^{*2}}, \; \rho_{s,0} = \frac{2}{\pi^2}\int_0^{k_{f_0}} p^2\, dp \frac{m_0^*}{\sqrt{p^2 + m_0^{*2}}}$$

$$\rho'_{s,0} = \frac{2}{\pi^2}\int_0^{k_{f_0}} p^2\, dp \left[m_0^{*2} \Big/ \left(\sqrt{p^2 + m_0^{*2}}\right)^3 - \frac{1}{\sqrt{p^2 + m_0^{*2}}} \right]$$

$$c_\omega^2 = \frac{1}{\rho_0}\left[(B/A)_0 + m - \sqrt{k_{f_0}^2 + m_0^{*2}}\right],$$

$$V(\Phi_0) = \left[(B/A)_0 + m\right]\rho_0 - \langle E \rangle_{k_{f_0}} - \frac{1}{2}c_\omega^2\rho_0^2, \; V'(\Phi_0) = \rho_{s,0}$$

Computer algorithm

$\sigma - \omega - \boldsymbol{\rho}$ model

- Input as constraints:

$$\rho_0, (B/A)_0, K_0, m_0^*, J, m, m_\sigma, m_\omega \text{ and } m_{\boldsymbol{\rho}}$$

$$\Phi_0 = m - m_0^*, \; k_{f_0} = \left(\frac{3\pi^2\rho_0}{2}\right)^{1/3}, \; \mu_0 = \sqrt{k_{f_0}^2 + m_0^{*2}}$$

$$\langle E \rangle_{k_{f_0}} = \frac{2}{\pi^2}\int_0^{k_{f_0}} p^2\,dp\sqrt{p^2 + m_0^{*2}}, \; \rho_{s,0} = \frac{2}{\pi^2}\int_0^{k_{f_0}} p^2\,dp\frac{m_0^*}{\sqrt{p^2 + m_0^{*2}}}$$

$$\rho'_{s,0} = \frac{2}{\pi^2}\int_0^{k_{f_0}} p^2\,dp\left[m_0^{*2}\Big/\left(\sqrt{p^2 + m_0^{*2}}\right)^3 - \frac{1}{\sqrt{p^2 + m_0^{*2}}}\right]$$

$$c_\omega^2 = \frac{1}{\rho_0}\left[(B/A)_0 + m - \sqrt{k_{f_0}^2 + m_0^{*2}}\right],$$

$$V(\Phi_0) = \left[(B/A)_0 + m\right]\rho_0 - \langle E \rangle_{k_{f_0}} - \frac{1}{2}c_\omega^2\rho_0^2, \; V'(\Phi_0) = \rho_{s,0}$$

$$V''(\Phi_0) = \rho'_{s,0} + \frac{(m_0^*/\mu_0)^2}{\frac{\pi^2}{2k_{f_0}\mu_0} + c_\omega^2 - \frac{K_0}{9\rho_0}}$$

Computer algorithm

$\sigma - \omega - \rho$ model

Computer algorithm

$\sigma - \omega - \rho$ model

- The parameters of the Lagrangian are found as

$$\begin{aligned}
\frac{1}{c_\sigma^2} &= \frac{2}{\Phi_0^2}\left[6\,V(\Phi_0) - 3\Phi_0\,V'(\Phi_0) + \frac{1}{2}\Phi_0^2\,V''(\Phi_0)\right] \\
\overline{g_2} &= \frac{3}{\Phi_0^3}\left[-8\,V(\Phi_0) + 5\Phi_0\,V'(\Phi_0) - \Phi_0^2\,V''(\Phi_0)\right] \\
\overline{g_3} &= \frac{4}{\Phi_0^4}\left[3\,V(\Phi_0) - 2\Phi_0\,V'(\Phi_0) + \frac{1}{2}\Phi_0^2\,V''(\Phi_0)\right]
\end{aligned}$$

Computer algorithm

$\sigma - \omega - \boldsymbol{\rho}$ model

- The parameters of the Lagrangian are found as

$$
\begin{aligned}
\frac{1}{c_\sigma^2} &= \frac{2}{\Phi_0^2}\left[6\,V(\Phi_0) - 3\Phi_0\,V'(\Phi_0) + \frac{1}{2}\Phi_0^2\,V''(\Phi_0)\right] \\
\overline{\mathrm{g}_2} &= \frac{3}{\Phi_0^3}\left[-8\,V(\Phi_0) + 5\Phi_0\,V'(\Phi_0) - \Phi_0^2\,V''(\Phi_0)\right] \\
\overline{\mathrm{g}_3} &= \frac{4}{\Phi_0^4}\left[3\,V(\Phi_0) - 2\Phi_0\,V'(\Phi_0) + \frac{1}{2}\Phi_0^2\,V''(\Phi_0)\right] \\
\Rightarrow \mathrm{g}_\sigma &= m_\sigma\,c_\sigma,\ \ \mathrm{g}_\omega = m_\omega\,c_\omega,\ \ \mathrm{g}_2 = \overline{\mathrm{g}_2}\mathrm{g}_\omega^3,\ \ \mathrm{g}_3 = \overline{\mathrm{g}_2}\mathrm{g}_\omega^4 \\
\frac{\mathrm{g}_{\boldsymbol{\rho}}}{2} &= \left[\left(J - k_{f_0}^2 \Big/ 6\sqrt{k_{f_0}^2 + m_0^{*2}}\right)\frac{3\pi^2}{k_{f_0}^3}\,m_{\boldsymbol{\rho}}^2\right]^{1/2}
\end{aligned}
$$

Computer algorithm

$\sigma - \omega - \boldsymbol{\rho}$ model

- The parameters of the Lagrangian are found as

$$
\begin{aligned}
\frac{1}{c_\sigma^2} &= \frac{2}{\Phi_0^2}\left[6\,V(\Phi_0) - 3\Phi_0\,V'(\Phi_0) + \frac{1}{2}\Phi_0^2\,V''(\Phi_0)\right] \\
\overline{\mathrm{g}_2} &= \frac{3}{\Phi_0^3}\left[-8\,V(\Phi_0) + 5\Phi_0\,V'(\Phi_0) - \Phi_0^2\,V''(\Phi_0)\right] \\
\overline{\mathrm{g}_3} &= \frac{4}{\Phi_0^4}\left[3\,V(\Phi_0) - 2\Phi_0\,V'(\Phi_0) + \frac{1}{2}\Phi_0^2\,V''(\Phi_0)\right] \\
\Rightarrow \mathrm{g}_\sigma &= m_\sigma\,c_\sigma,\ \ \mathrm{g}_\omega = m_\omega\,c_\omega,\ \ \mathrm{g}_2 = \overline{\mathrm{g}_2}\mathrm{g}_\omega^3,\ \ \mathrm{g}_3 = \overline{\mathrm{g}_2}\mathrm{g}_\omega^4 \\
\frac{\mathrm{g}_{\boldsymbol{\rho}}}{2} &= \left[\left(J - k_{f_0}^2 \Big/ 6\sqrt{k_{f_0}^2 + m_0^{*2}}\right)\frac{3\pi^2}{k_{f_0}^3}\,m_{\boldsymbol{\rho}}^2\right]^{1/2}
\end{aligned}
$$

- **Assignment: Find the parameters of $\sigma - \omega - \boldsymbol{\rho}$ model for $\rho_0 = 0.16\ \mathrm{fm}^{-3}, (\boldsymbol{B}/\boldsymbol{A})_0 = -16.0\ \mathrm{MeV}, \boldsymbol{K}_0 = 230\ \mathrm{MeV}, \boldsymbol{m}_0^* = 0.6, \boldsymbol{J} = 32.0\ \mathrm{MeV}, \boldsymbol{m} = 939\ \mathrm{MeV}, \boldsymbol{m}_\sigma = 600\ \mathrm{MeV}, \boldsymbol{m}_\omega = 782\ \mathrm{MeV}$ and $\boldsymbol{m}_{\boldsymbol{\rho}} = 763\ \mathrm{MeV}$. Set up the self-consistent mean field solutions to find ϵ and p as a function of density.**

References:

- Compact Stars, Springer, **Norman K. Glendenning**.
- Quantum Field Theory, Cambridge University Press, **Lewis H. Ryder**.